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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Laws of Motion

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	140 questions
Numerical	25 questions
Answer key	at the end

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1 **Solve**
Print it, or work through it on screen.

2 **Check**
The answer key is on the last pages.

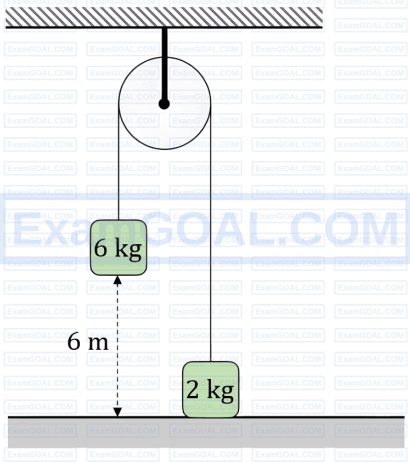
3 **Understand**
Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 140 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6 m above the ground is ____ $\mathrm{m/s}$. Assume pulley is massless and string is light and inextensible. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

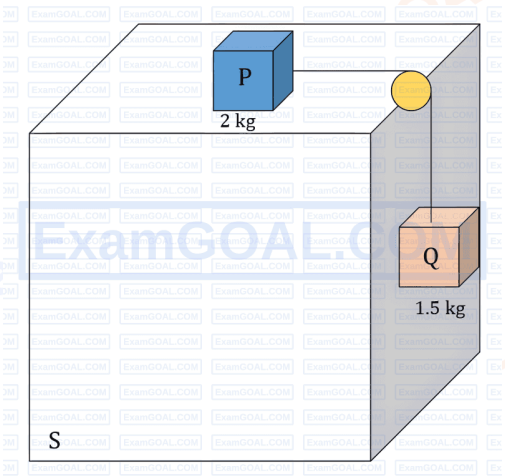


(A) 7.74 (B) 7.20 (C) 6.55 (D) 4.50

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pulley which is fixed on the edge of a cube (S), as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ. The cube (S) moves towards the right with acceleration of $\frac{g}{2}$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is ____ (take $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) 0.33 (B) 0.67 (C) 1 (D) 0.5

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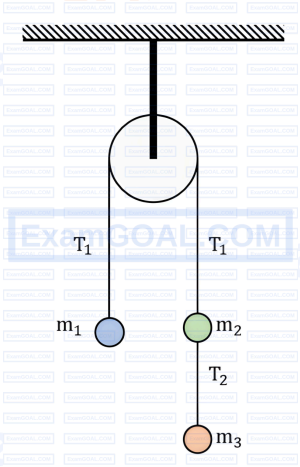
Q3. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Three masses $m_1 = 4\,\text{kg}, m_2 = 4\,\text{kg}$ and $m_3 = 6\,\text{kg}$ are suspended from a fixed smooth frictionless pulley as shown in the figure below. The value of T_1/T_2 is

____

(take $g = 10\,\text{m/s}^2$)



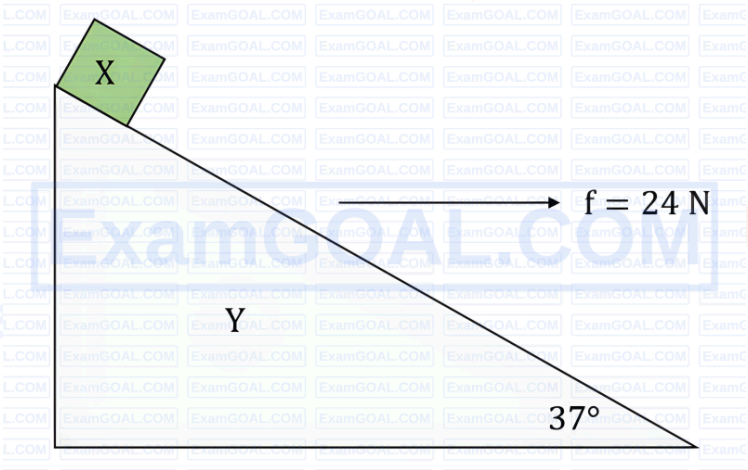
(A) 5/3 (B) 2/3 (C) 3/5 (D) 2/5

Q4. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

A wedge Y with mass of 10 kg and all frictionless surfaces and the inclined surface making 37° with horizontal. A block X with mass 2 kg is placed at the highest point of the wedge as shown in figure is at rest. At $t = 0$ wedge (Y) is pulled toward right with constant force (f) of 24 N . Taking the block X at rest at $t = 0$, the time taken by it to slide down 8.8 m on the slope, while Y is on the move, is ____ s.

(take $\tan(37^\circ) = 3/4$ and $\text{g} = 10\,\text{m/s}^2$)



(A) 2 (B) 4 (C) $\sqrt{2}$ (D) $2\sqrt{2}$

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Q5. JEE Main 2026 (Online) 4th April Evening Shift

At $t = 0$, a body of mass 100 g starts moving under the influence of a force $(5\hat{i} + 10\hat{j})$N. After 2 s its position is $(2x\hat{i} + 5y\hat{j})$m. The ratio $x : y$ is ____ .

(A) $1 : 2$ (B) $2 : 5$ (C) $5 : 2$ (D) $5 : 4$

Open solution

Q6. JEE Main 2026 (Online) 4th April Morning Shift

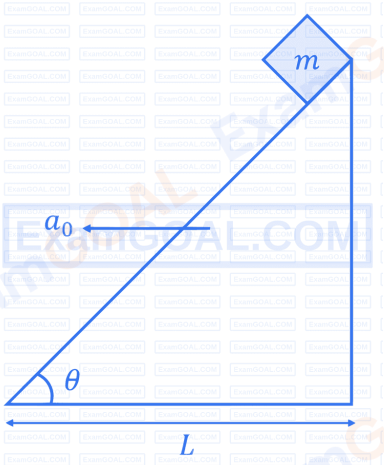
The time taken by a block of mass m to slide down from the highest point to the lowest point on a rough inclined plane is 50% more compared to the time taken by the same block on identical inclined smooth plane. Both inclined planes are at 45° with the horizontal. The coefficient of kinetic friction between the rough inclined surface and block is ____

(A) 3/4 (B) 2/3 (C) 5/9 (D) 4/9

Open solution

Q7. JEE Main 2026 (Online) 28th January Evening Shift

A small block of mass **m** slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0. The angle between the inclined plane and ground is θ and its base length is L. Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is _____.



(A) $\sqrt{\dfrac{2L}{g\sin\theta - a_0\cos\theta}}$ (B) $\sqrt{\dfrac{4L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$

(C) $\sqrt{\dfrac{2L}{g\sin 2\theta - a_0(1+\cos 2\theta)}}$ (D) $\sqrt{\dfrac{4L}{g\cos^2\theta - a_0\sin\theta\cos\theta}}$

Open solution

Q8. JEE Main 2026 (Online) 28th January Morning Shift

A block of mass 5 kg is moving on an inclined plane which makes an angle of 30° with the horizontal. Friction coefficient between the block and inclined plane surface is $\frac{\sqrt{3}}{2}$. The force to be applied on the block so that the block will move down without acceleration is ____ N.

$(g = 10\,\text{m/s}^2)$.

(A) 25 (B) 12.5 (C) 15 (D) 7.5

Open solution

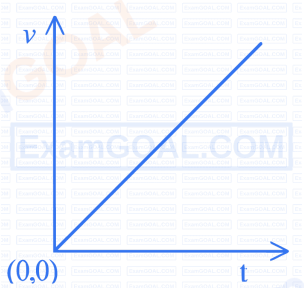
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Q9. JEE Main 2026 (Online) 28th January Morning Shift

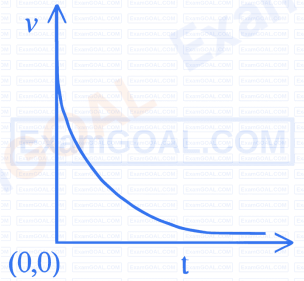
Open solution

A particle of mass m falls from rest through a resistive medium having resistive force, $F = -kv$, where v is the velocity of the particle and k is a constant. Which of the following graphs represents velocity (v) versus time (t) ?

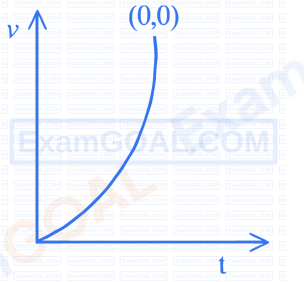
(A)



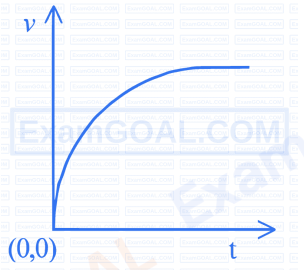
(B)



(C)



(D)



Q10. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30°. Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is ____ .

(A) $m\,\mathrm{g}$ (B) $\frac{\sqrt{3}}{2}mg$ (C) $\frac{1}{2}mg$ (D) $\sqrt{3}m\,\mathrm{g}$

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Q11. JEE Main 2026 (Online) 23rd January Evening Shift

A block is sliding down with constant velocity on an inclined plane of slope θ and at an instant $t = 0$ this block is given an upward momentum so that it starts moving up on the inclined surface with velocity u. The distance (S) travelled by the block before its velocity become zero, is ____ .

(g = gravitational acceleration)

(A) $\frac{2u^2}{g\cos\theta}$

(B) $\frac{u^2}{\sqrt{2}g\cos\theta}$

(C) $\frac{u^2}{4g\sin\theta}$

(D) $\frac{u^2}{2g\cos\theta}$

Open solution

Q12. JEE Main 2026 (Online) 21st January Morning Shift

A 4 kg mass moves under the influence of a force $\vec{F} = \left(4t^3\hat{i} - 3t\hat{j}\right)\text{N}$ where t is the time in second. If mass starts from origin at $t = 0$, the velocity and position after $t = 2\,\text{s}$ will be:

(A) $\vec{v} = 3\hat{i} + \frac{3}{2}\hat{j} \quad \vec{r} = \frac{6}{5}\hat{i} + \hat{j}$

(B) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j} \quad \vec{r} = \frac{6}{5}\hat{i} - \hat{j}$

(C) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j} \quad \vec{r} = \frac{8}{5}\hat{i} - \hat{j}$

(D) $\vec{v} = 4\hat{i} + \frac{5}{2}\hat{j} \quad \vec{r} = \frac{8}{5}\hat{i} + 2\hat{j}$

Open solution

Q13. JEE Main 2025 (Online) 8th April Evening Shift

A body of mass 2 kg moving with velocity of $\vec{v}_{in} = 3\hat{i} + 4\hat{j}\,\text{ms}^{-1}$ enters into a constant force field of 6N directed along positive z-axis. If the body remains in the field for a period of $\frac{5}{3}$ seconds, then velocity of the body when it emerges from force field is.

(A) $3\hat{i} + 4\hat{j} + \sqrt{5}\hat{k}$

(B) $4\hat{i} + 3\hat{j} + 5\hat{k}$

(C) $3\hat{i} + 4\hat{j} - 5\hat{k}$

(D) $3\hat{i} + 4\hat{j} + 5\hat{k}$

Open solution

Q14. JEE Main 2025 (Online) 7th April Evening Shift

An object with mass 500 g moves along x-axis with speed $v = 4\sqrt{x}$ m/s. The force acting on the object is :

(A) 8 N

(B) 4 N

(C) 5 N

(D) 6 N

Open solution

Q15. JEE Main 2025 (Online) 7th April Morning Shift

A cubic block of mass m is sliding down on an inclined plane at 60° with an acceleration of $\frac{g}{2}$, the value of coefficient of kinetic friction is

(A) $\frac{\sqrt{3}}{2}$

(B) $\frac{\sqrt{2}}{3}$

(C) $1 - \frac{\sqrt{3}}{2}$

(D) $\sqrt{3} - 1$

Open solution

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Q16. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

A body of mass m is suspended by two strings making angles θ_1 and θ_2 with the horizontal ceiling with tensions T_1 and T_2 simultaneously. T_1 and T_2 are related by $T_1 = \sqrt{3}T_2$, the angles θ_1 and θ_2 are

(A) $\theta_1 = 30^\circ \theta_2 = 60^\circ$ with $T_2 = \frac{3\text{mg}}{4}$

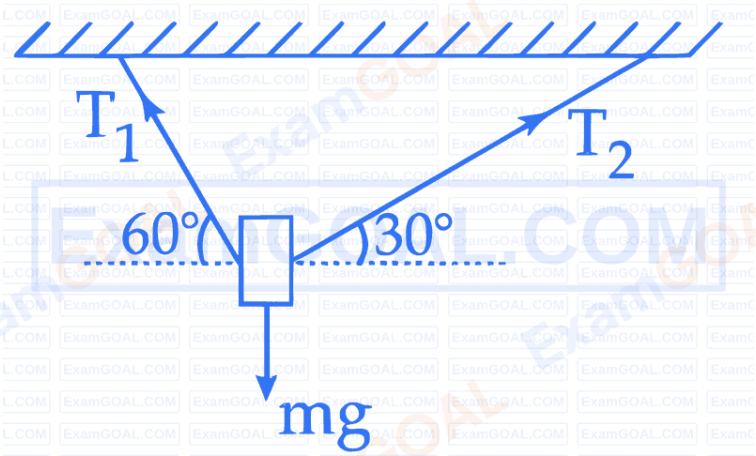
(B) $\theta_1 = 45^\circ \theta_2 = 45^\circ$ with $T_2 = \frac{3mg}{4}$

(C) $\theta_1 = 30^\circ \theta_2 = 60^\circ$ with $T_2 = \frac{4mg}{5}$

(D) $\theta_1 = 60^\circ \theta_2 = 30^\circ$ with $T_2 = \frac{mg}{2}$

Q17. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution



A body of mass 1 kg is suspended with the help of two strings making angles as shown in figure. Magnitudes of tensions T_1 and T_2, respectively, are (in N) :

(Take acceleration due to gravity $10\ \mathrm{m/s^2}$)

(A) $5\sqrt{3}, 5$

(B) $5\sqrt{3}, 5\sqrt{3}$

(C) 5,5

(D) $5, 5\sqrt{3}$

Q18. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

A balloon and its content having mass M is moving up with an acceleration 'a'. The mass that must be released from the content so that the balloon starts moving up with an acceleration '3a' will be

(Take 'g' as acceleration due to gravity)

(A) $\frac{3Ma}{2a+g}$

(B) $\frac{2Ma}{3a+g}$

(C) $\frac{3Ma}{2a-g}$

(D) $\frac{2Ma}{3a-g}$

Q19. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

A massless spring gets elongated by amount x_1 under a tension of 5 N . Its elongation is x_2 under the tension of 7 N . For the elongation of $(5x_1 - 2x_2)$, the tension in the spring will be,

(A) 20 N

(B) 39 N

(C) 11 N

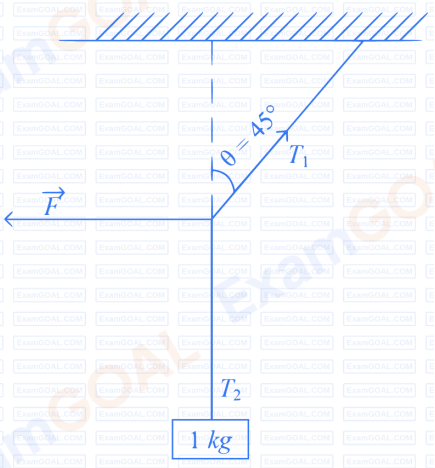
(D) 15 N

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Q20. JEE Main 2024 (Online) 9th April Evening Shift

A $1\ \mathrm{kg}$ mass is suspended from the ceiling by a rope of length $4\ \mathrm{m}$. A horizontal force 'F' is applied at the mid point of the rope so that the rope makes an angle of 45° with respect to the vertical axis as shown in figure. The magnitude of F is :

(Assume that the system is in equilibrium and $g = 10\ \mathrm{m/s^2}$)



Open solution

(A) 1 N (B) 10 N (C) $\frac{10}{\sqrt{2}}N$ (D) $\frac{1}{10\times\sqrt{2}}N$

Q21. JEE Main 2024 (Online) 9th April Morning Shift

A light unstretchable string passing over a smooth light pulley connects two blocks of masses m_1 and m_2. If the acceleration of the system is $\frac{g}{8}$, then the ratio of the masses $\frac{m_2}{m_1}$ is :

Open solution

(A) $5:3$ (B) $8:1$ (C) $9:7$ (D) $4:3$

Q22. JEE Main 2024 (Online) 8th April Evening Shift

A given object takes n times the time to slide down 45° rough inclined plane as it takes the time to slide down an identical perfectly smooth 45° inclined plane. The coefficient of kinetic friction between the object and the surface of inclined plane is :

Open solution

(A) $1-\frac{1}{\mathrm{n}^2}$ (B) $\sqrt{1-\frac{1}{\mathrm{n}^2}}$ (C) $1-n^2$ (D) $\sqrt{1-n^2}$

Q23. JEE Main 2024 (Online) 8th April Morning Shift

A player caught a cricket ball of mass $150\ \mathrm{g}$ moving at a speed of $20\ \mathrm{m/s}$. If the catching process is completed in $0.1\ \mathrm{s}$, the magnitude of force exerted by the ball on the hand of the player is:

Open solution

(A) 150 N (B) 3 N (C) 30 N (D) 300 N

Q24. JEE Main 2024 (Online) 6th April Evening Shift

A body of weight $200\ \mathrm{N}$ is suspended from a tree branch through a chain of mass $10\ \mathrm{kg}$. The branch pulls the chain by a force equal to (if $\mathrm{g} = 10\ \mathrm{m/s^2}$) :

Open solution

(A) 300 N (B) 100 N (C) 150 N (D) 200 N

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Q25. JEE Main 2024 (Online) 6th April Morning Shift

A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (where $m_2 > m_1$). If the acceleration of the system is $\frac{g}{\sqrt{2}}$, then the ratio of the masses $\frac{m_1}{m_2}$ is:

(A) $\frac{1+\sqrt{5}}{\sqrt{5}-1}$ (B) $\frac{\sqrt{3}+1}{\sqrt{2}-1}$ (C) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (D) $\frac{1+\sqrt{5}}{\sqrt{2}-1}$

Open solution

Q26. JEE Main 2024 (Online) 5th April Evening Shift

A particle moves in x-y plane under the influence of a force $\vec{F}$ such that its linear momentum is $\overrightarrow{\mathrm{p}}(\mathrm{t}) = \hat{i}\cos(\mathrm{kt}) - \hat{j}\sin(\mathrm{kt})$. If k is constant, the angle between $\overrightarrow{\mathrm{F}}$ and $\overrightarrow{\mathrm{p}}$ will be :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

Open solution

Q27. JEE Main 2024 (Online) 5th April Evening Shift

A heavy box of mass $50\ \mathrm{kg}$ is moving on a horizontal surface. If co-efficient of kinetic friction between the box and horizontal surface is 0.3 then force of kinetic friction is :

(A) 1470 N (B) 147 N (C) 1.47 N (D) 14.7 N

Open solution

Q28. JEE Main 2024 (Online) 5th April Morning Shift

A wooden block of mass $5\ \mathrm{kg}$ rests on a soft horizontal floor. When an iron cylinder of mass $25\ \mathrm{kg}$ is placed on the top of the block, the floor yields and the block and the cylinder together go down with an acceleration of $0.1\ \mathrm{ms}^{-2}$. The action force of the system on the floor is equal to :

(A) 297 N (B) 291 N (C) 196 N (D) 294 N

Open solution

Q29. JEE Main 2024 (Online) 4th April Evening Shift

A $2\ \mathrm{kg}$ brick begins to slide over a surface which is inclined at an angle of 45° with respect to horizontal axis. The co-efficient of static friction between their surfaces is:

(A) 1 (B) 1.7 (C) 0.5 (D) $\frac{1}{\sqrt{3}}$

Open solution

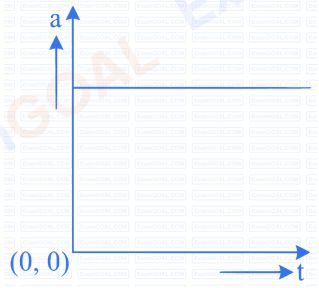
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Q30. JEE Main 2024 (Online) 4th April Morning Shift

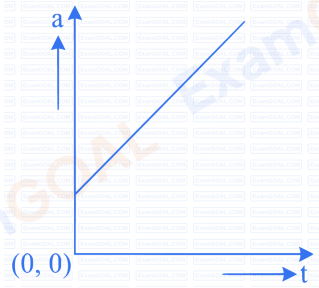
A wooden block, initially at rest on the ground, is pushed by a force which increases linearly with time t. Which of the following curve best describes acceleration of the block with time :

Open solution

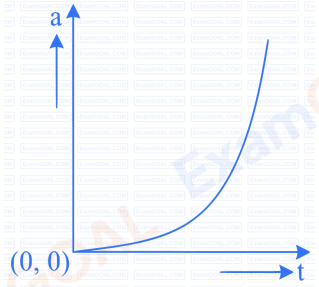
(A)



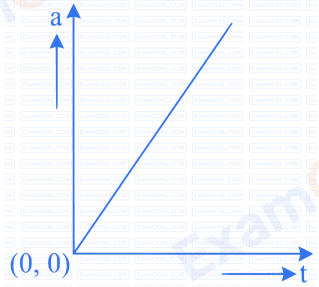
(B)



(C)



(D)



Q31. JEE Main 2024 (Online) 1st February Evening Shift

A body of mass $4\,\text{kg}$ experiences two forces $\vec{F}_1 = 5\hat{i} + 8\hat{j} + 7\hat{k}$ and $\vec{F}_2 = 3\hat{i} - 4\hat{j} - 3\hat{k}$. The acceleration acting on the body is :

Open solution

(A) $2\hat{i} + \hat{j} + \hat{k}$ (B) $4\hat{i} + 2\hat{j} + 2\hat{k}$ (C) $-2\hat{i} - \hat{j} - \hat{k}$ (D) $2\hat{i} + 3\hat{j} + 3\hat{k}$

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Q32. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

A cricket player catches a ball of mass $120\ \mathrm{g}$ moving with $25\ \mathrm{m/s}$ speed. If the catching process is completed in $0.1\ \mathrm{s}$ then the magnitude of force exerted by the ball on the hand of player will be (in SI unit) :

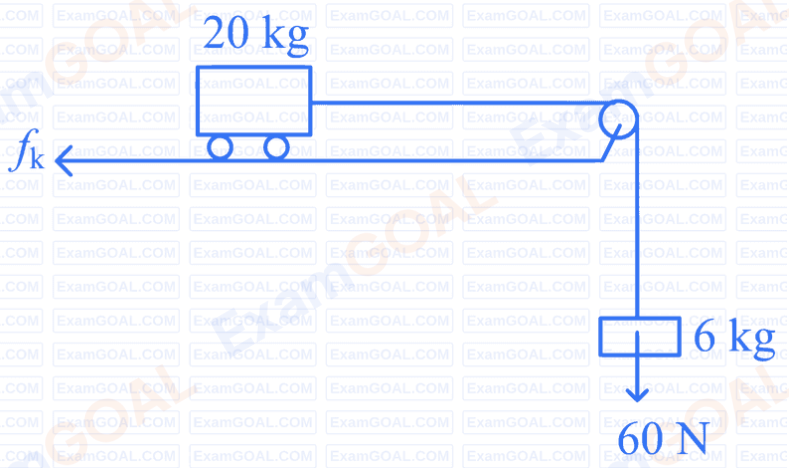
(A) 30 (B) 24 (C) 12 (D) 25

Q33. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

Consider a block and trolley system as shown in figure. If the coefficient of kinetic friction between the trolley and the surface is 0.04 , the acceleration of the system in ms^{-2} is :

(Consider that the string is massless and unstretchable and the pulley is also massless and frictionless) :

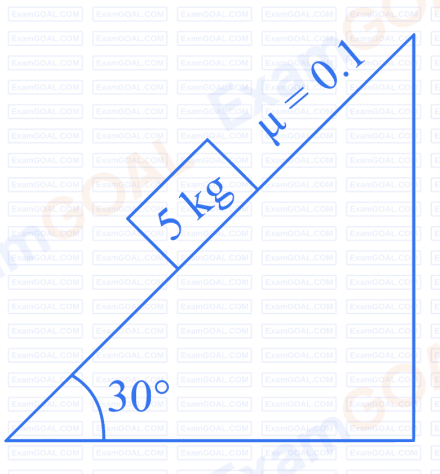


(A) 1.2 (B) 4 (C) 3 (D) 2

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Q34. JEE Main 2024 (Online) 31st January Evening Shift

Open solution



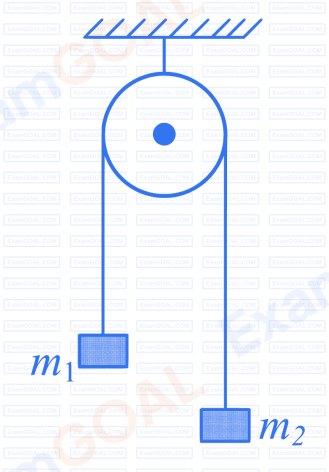
A block of mass $5\ \text{kg}$ is placed on a rough inclined surface as shown in the figure. If $\overrightarrow{F_1}$ is the force required to just move the block up the inclined plane and $\overrightarrow{F_2}$ is the force required to just prevent the block from sliding down, then the value of $\left|\overrightarrow{F_1}\right| - \left|\overrightarrow{F_2}\right|$ is : [Use $g = 10\ \text{m/s}^2$]

(A) $5\sqrt{3}N$ (B) $\frac{5\sqrt{3}}{2}N$ (C) $10\ \text{N}$ (D) $25\sqrt{3}N$

Q35. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

A light string passing over a smooth light fixed pulley connects two blocks of masses m_1 and m_2. If the acceleration of the system is $g/8$, then the ratio of masses is:



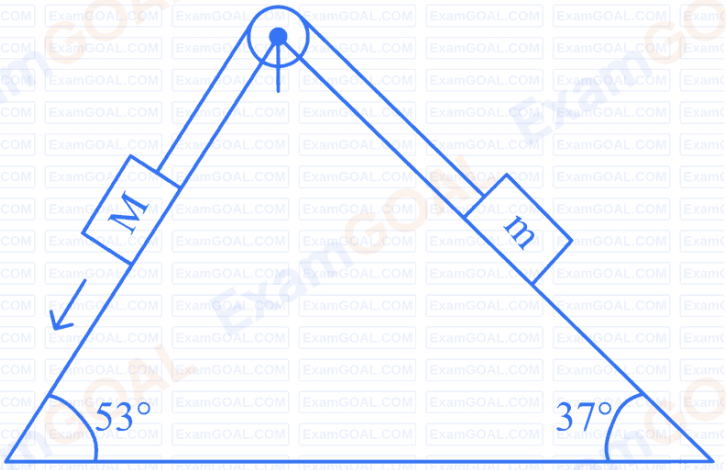
(A) $\frac{8}{1}$ (B) $\frac{9}{7}$ (C) $\frac{5}{3}$ (D) $\frac{4}{3}$

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Q36. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

In the given arrangement of a doubly inclined plane two blocks of masses M and m are placed. The blocks are connected by a light string passing over an ideal pulley as shown. The coefficient of friction between the surface of the plane and the blocks is 0.25. The value of m, for which $M = 10\,\text{kg}$ will move down with an acceleration of $2\,\text{m/s}^2$, is: (take $g = 10\,\text{m/s}^2$ and $\tan 37^\circ = 3/4$)



(A) 4.5 kg (B) 6.5 kg (C) 9 kg (D) 2.25 kg

Q37. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

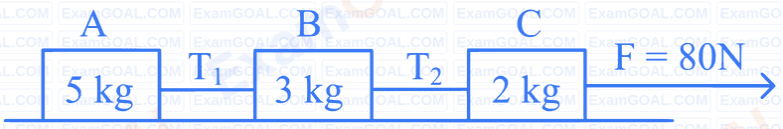
A block of mass m is placed on a surface having vertical cross-section given by $y = x^2/4$. If coefficient of friction is 0.5, the maximum height above the ground at which block can be placed without slipping is:

(A) 1/2 m (B) 1/3 m (C) 1/6 m (D) 1/4 m

Q38. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Three blocks A, B and C are pulled on a horizontal smooth surface by a force of $80\,\text{N}$ as shown in figure



The tensions T_1 and T_2 in the string are respectively :

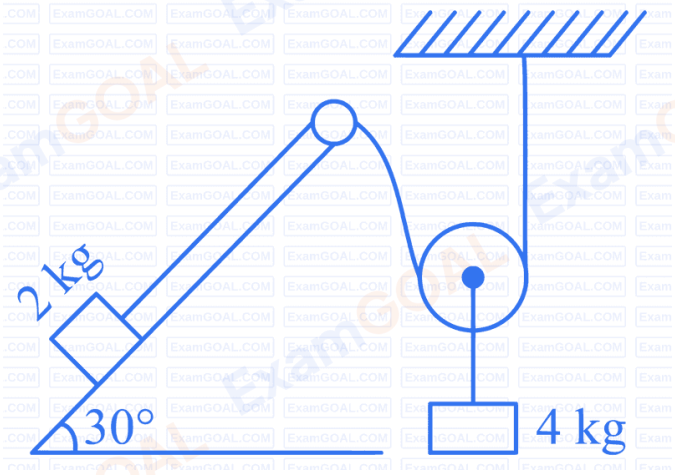
(A) 40N, 64N (B) 60N, 80N (C) 80N, 100N (D) 88N, 96N

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Q39. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

All surfaces shown in figure are assumed to be frictionless and the pulleys and the string are light. The acceleration of the block of mass $2\ \text{kg}$ is :



(A) $\frac{g}{2}$ (B) $\frac{g}{4}$ (C) g (D) $\frac{g}{3}$

Q40. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

Given below are two statements :

Statement (I) : The limiting force of static friction depends on the area of contact and independent of materials.

Statement (II) : The limiting force of kinetic friction is independent of the area of contact and depends on materials.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q41. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

The position vector of a particle related to time t is given by

$$\vec{r} = \left(10t\hat{i} + 15t^2\hat{j} + 7\hat{k}\right)m$$

The direction of net force experienced by the particle is :

(A) Positive x - axis (B) Positive y - axis

(C) Positive z - axis (D) In x - y plane

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Q42. JEE Main 2023 (Online) 12th April Morning Shift

Three forces $F_1 = 10\,\text{N}, F_2 = 8\,\text{N}, F_3 = 6\,\text{N}$ are acting on a particle of mass $5\,\text{kg}$. The forces F_2 and F_3 are applied perpendicularly so that particle remains at rest. If the force F_1 is removed, then the acceleration of the particle is:

Open solution

(A) $4.8\,\text{ms}^{-2}$ (B) $7\,\text{ms}^{-2}$ (C) $2\,\text{ms}^{-2}$ (D) $0.5\,\text{ms}^{-2}$

Q43. JEE Main 2023 (Online) 11th April Evening Shift

A body of mass $500\,\text{g}$ moves along x-axis such that its velocity varies with displacement x according to the relation $v = 10\sqrt{x}\,\text{m/s}$ the force acting on the body is:-

Open solution

(A) 166 N (B) 5 N (C) 25 N (D) 125 N

Q44. JEE Main 2023 (Online) 8th April Morning Shift

At any instant the velocity of a particle of mass $500\,\text{g}$ is $\left(2t\hat{i} + 3t^2\hat{j}\right)\text{ms}^{-1}$. If the force acting on the particle at $t = 1\,\text{s}$ is $(\hat{i} + x\hat{j})\text{N}$. Then the value of x will be:

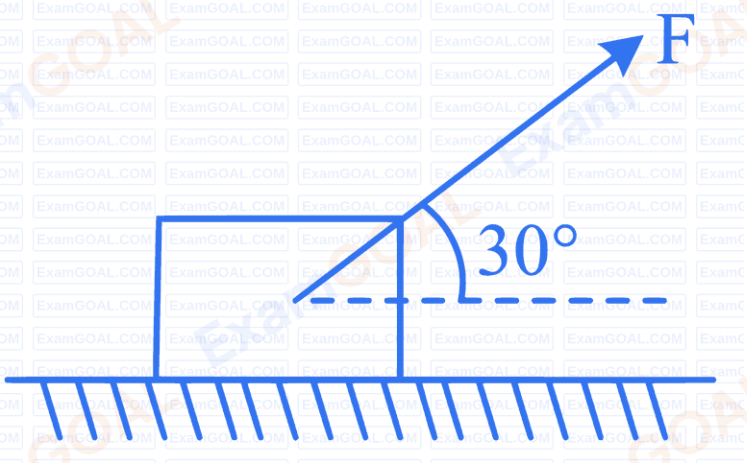
Open solution

(A) 2 (B) 4 (C) 6 (D) 3

Q45. JEE Main 2023 (Online) 1st February Evening Shift

As shown in the figure a block of mass 10 kg lying on a horizontal surface is pulled by a force F acting at an angle $30°$, with horizontal. For $\mu_s = 0.25$, the block will just start to move for the value of F : [Given $g = 10\,\text{ms}^{-2}$]

Open solution



(A) 25.2 N (B) 35.7 N (C) 20 N (D) 33.3 N

Q46. JEE Main 2023 (Online) 1st February Morning Shift

A block of mass $5\,\text{kg}$ is placed at rest on a table of rough surface. Now, if a force of $30\,\text{N}$ is applied in the direction parallel to surface of the table, the block slides through a distance of $50\,\text{m}$ in an interval of time $10\,\text{s}$. Coefficient of kinetic friction is (given, $g = 10\,\text{ms}^{-2}$):

Open solution

(A) 0.25 (B) 0.75 (C) 0.60 (D) 0.50

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Q47. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

A body of mass $10\ \text{kg}$ is moving with an initial speed of $20\ \text{m/s}$. The body stops after $5\ \text{s}$ due to friction between body and the floor. The value of the coefficient of friction is:

(Take acceleration due to gravity $g = 10\ \text{ms}^{-2}$)

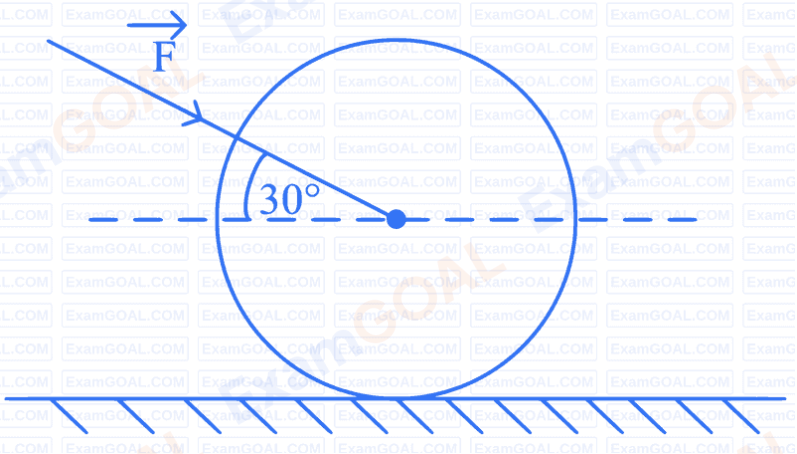
(A) 0.3 (B) 0.2 (C) 0.5 (D) 0.4

Q48. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

As shown in figure, a $70\ \text{kg}$ garden roller is pushed with a force of $\vec{F} = 200\ \text{N}$ at an angle of 30° with horizontal. The normal reaction on the roller is

(Given $\text{g} = 10\ \text{m}\,\text{s}^{-2}$)



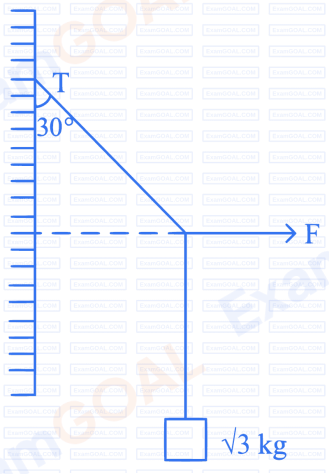
(A) 800 N (B) 600 N (C) $200\sqrt{3}$ N (D) $800\sqrt{2}$ N

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Q49. JEE Main 2023 (Online) 30th January Evening Shift

A block of $\sqrt{3}\,\mathrm{kg}$ is attached to a string whose other end is attached to the wall. An unknown force F is applied so that the string makes an angle of 30° with the wall. The tension T is: (Given $\mathrm{g} = 10\,\mathrm{ms}^{-2}$)

Open solution



(A) 15 N (B) 10 N (C) 25 N (D) 20 N

Q50. JEE Main 2023 (Online) 29th January Evening Shift

A force acts for 20 s on a body of mass 20 kg, starting from rest, after which the force ceases and then body describes 50 m in the next 10 s. The value of force will be:

(A) 40 N (B) 20 N (C) 5 N (D) 10 N

Open solution

Q51. JEE Main 2023 (Online) 29th January Evening Shift

The time taken by an object to slide down 45° rough inclined plane is n times as it takes to slide down a perfectly smooth 45° incline plane. The coefficient of kinetic friction between the object and the incline plane is :

(A) $1-\frac{1}{n^2}$ (B) $1+\frac{1}{n^2}$ (C) $\sqrt{1-\frac{1}{n^2}}$ (D) $\sqrt{\frac{1}{1-n^2}}$

Open solution

Q52. JEE Main 2023 (Online) 29th January Morning Shift

A block of mass m slides down the plane inclined at angle 30° with an acceleration $\frac{g}{4}$. The value of coefficient of kinetic friction will be:

(A) $\frac{2\sqrt{3}-1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2\sqrt{3}}$ (D) $\frac{2\sqrt{3}+1}{2}$

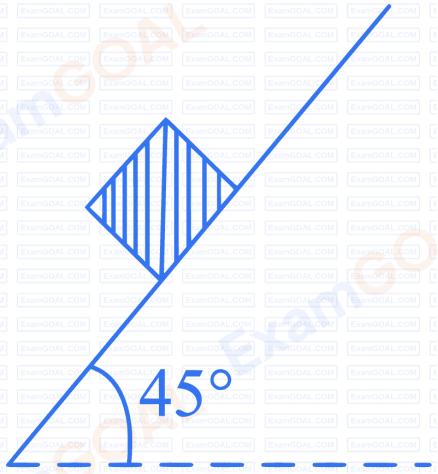
Open solution

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Q53. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Consider a block kept on an inclined plane (incline at 45°) as shown in the figure. If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down, the coefficient of friction between the block and inclined plane(μ) is equal to :

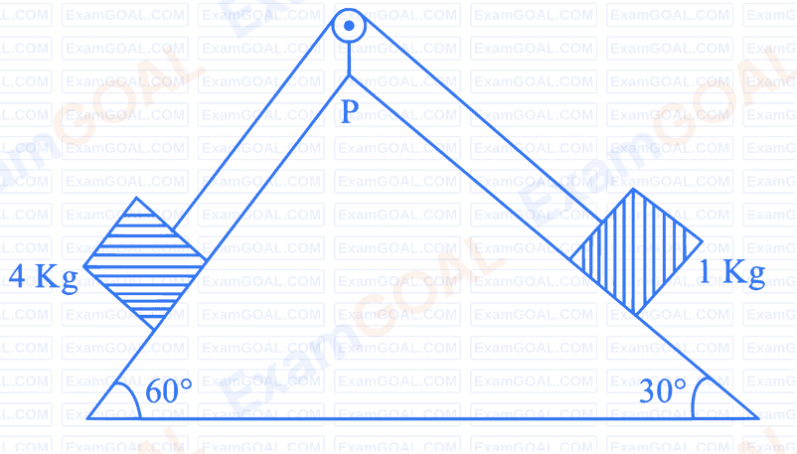


(A) 0.60 (B) 0.33 (C) 0.25 (D) 0.50

Q54. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

As per given figure, a weightless pulley P is attached on a double inclined frictionless surfaces. The tension in the string (massless) will be (if g = 10 m/s^2)



(A) $\left(4\sqrt{3}-1\right)N$ (B) $\left(4\sqrt{3}+1\right)N$ (C) $4\left(\sqrt{3}-1\right)N$ (D) $4\left(\sqrt{3}+1\right)N$

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Q55. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Given below are two statements :

Statement I : An elevator can go up or down with uniform speed when its weight is balanced with the tension of its cable.

Statement II : Force exerted by the floor of an elevator on the foot of a person standing on it is more than his/her weight when the elevator goes down with increasing speed.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q56. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

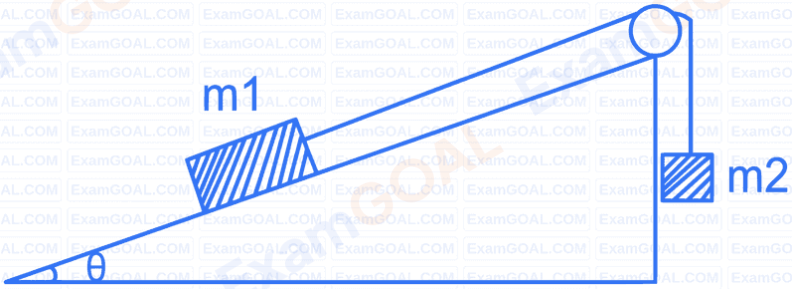
A bullet of mass 10 gram is fired with velocity of $100\ \mathrm{m/s}$ from the gun of mass $1\ \mathrm{kg}$. Recoil speed of the gun is :

(A) $1\ \mathrm{m/s}$ (B) $10\ \mathrm{m/s}$ (C) $0.5\ \mathrm{m/s}$ (D) $0.1\ \mathrm{m/s}$

Q57. JEE Main 2022 (Online) 29th July Evening Shift

Open solution

Two bodies of masses $m_1 = 5\ \mathrm{kg}$ and $m_2 = 3\ \mathrm{kg}$ are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass $\mathrm{m_1}$ will be : [Take $\mathrm{g} = 10\ \mathrm{ms}^{-2}$]



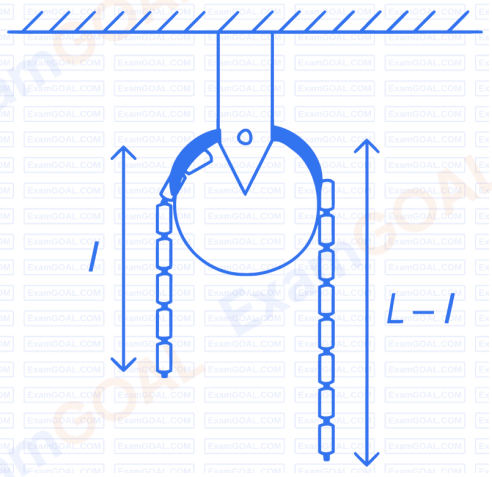
(A) 30 N (B) 40 N (C) 50 N (D) 60 N

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Q58. JEE Main 2022 (Online) 28th July Evening Shift

A uniform metal chain of mass m and length 'L' passes over a massless and frictionless pulley. It is released from rest with a part of its length 'l' hanging on one side and rest of its length '$L - l$' is hanging on the other side of the pulley. At a certain point of time, when $l = \frac{L}{x}$, the acceleration of the chain is $\frac{g}{2}$. The value of x is _____.

Open solution



(A) 6 (B) 2 (C) 1.5 (D) 4

Q59. JEE Main 2022 (Online) 28th July Morning Shift

A balloon has mass of $10\ \mathrm{g}$ in air. The air escapes from the balloon at a uniform rate with velocity $4.5\ \mathrm{cm/s}$. If the balloon shrinks in $5\ \mathrm{s}$ completely. Then, the average force acting on that balloon will be (in dyne).

Open solution

(A) 3 (B) 9 (C) 12 (D) 18

Q60. JEE Main 2022 (Online) 27th July Evening Shift

A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ. The magnitude of the contact force will be :

Open solution

(A) Mg (B) $\mathrm{Mg}\cos\theta$

(C) $\sqrt{\mathrm{Mg}\sin\theta + \mathrm{Mg}\cos\theta}$ (D) $\mathrm{Mg}\sin\theta\sqrt{1+\mu}$

Q61. JEE Main 2022 (Online) 27th July Evening Shift

A block 'A' takes 2 s to slide down a frictionless incline of 30° and length 'l', kept inside a lift going up with uniform velocity 'v'. If the incline is changed to 45°, the time taken by the block, to slide down the incline, will be approximately :

Open solution

(A) 2.66 s (B) 0.83 s (C) 1.68 s (D) 0.70 s

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Q62. JEE Main 2022 (Online) 27th July Morning Shift

A bag is gently dropped on a conveyor belt moving at a speed of $2\ \mathrm{m/s}$. The coefficient of friction between the conveyor belt and bag is 0.4. Initially, the bag slips on the belt before it stops due to friction. The distance travelled by the bag on the belt during slipping motion, is : [Take $\mathrm{g} = 10\ \mathrm{m/s^2}$]

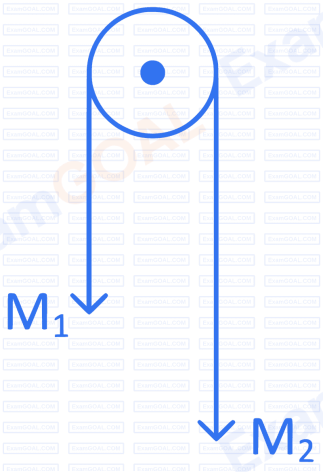
Open solution

(A) 2 m (B) 0.5 m (C) 3.2 m (D) 0.8 m

Q63. JEE Main 2022 (Online) 26th July Evening Shift

Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1, the acceleration of the system is a_1. When the mass M_2 is thrice that of M_1, the acceleration of the system is a_2. The ratio $\frac{a_1}{a_2}$ will be :

Open solution



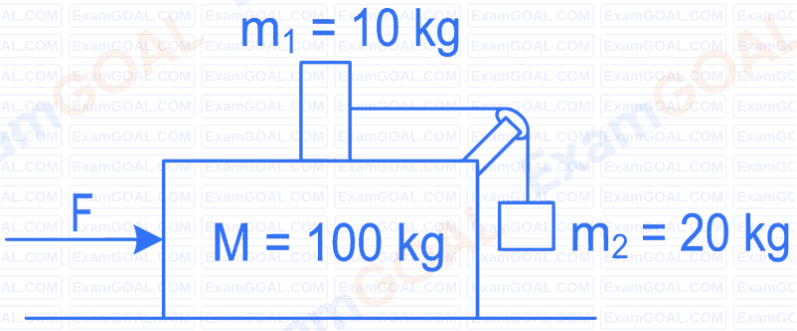
(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{1}{2}$

Q64. JEE Main 2022 (Online) 26th July Morning Shift

Three masses $M = 100\ \mathrm{kg}$, $\mathrm{m_1} = 10\ \mathrm{kg}$ and $\mathrm{m_2} = 20\ \mathrm{kg}$ are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force F is applied on the system so that the mass $\mathrm{m_2}$ moves upward with an acceleration of $2\ \mathrm{ms^{-2}}$. The value of F is :

(Take $\mathrm{g} = 10\ \mathrm{ms^{-2}}$)

Open solution



(A) 3360 N (B) 3380 N (C) 3120 N (D) 3240 N

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Q65. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

A monkey of mass $50\ \text{kg}$ climbs on a rope which can withstand the tension (T) of $350\ \text{N}$. If monkey initially climbs down with an acceleration of $4\ \text{m/s}^2$ and then climbs up with an acceleration of $5\ \text{m/s}^2$. Choose the correct option $(g = 10\ \text{m/s}^2)$.

(A) $T = 700\ \text{N}$ while climbing upward

(B) $T = 350\ \text{N}$ while going downward

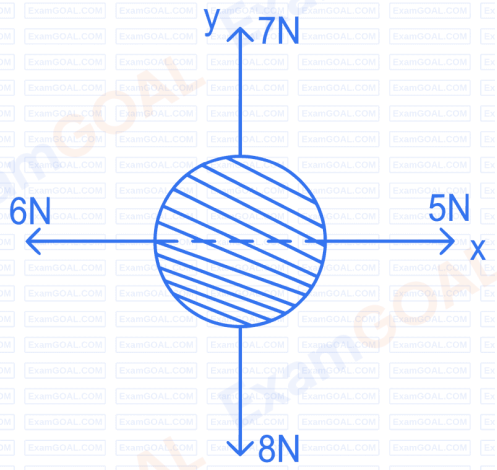
(C) Rope will break while climbing upward

(D) Rope will break while going downward

Q66. JEE Main 2022 (Online) 25th July Evening Shift

Open solution

For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero?



(A) $\sqrt{2}N, 45^\circ$

(B) $\sqrt{2}N, 135^\circ$

(C) $\frac{2}{\sqrt{3}}N, 30^\circ$

(D) $2N, 45^\circ$

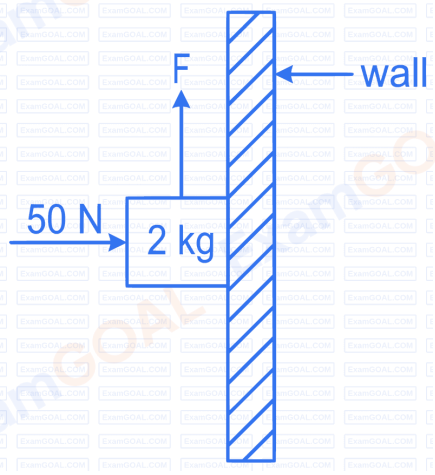
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Q67. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

A 2 kg block is pushed against a vertical wall by applying a horizontal force of 50 N. The coefficient of static friction between the block and the wall is 0.5. A force F is also applied on the block vertically upward (as shown in figure). The maximum value of F applied, so that the block does not move upward, will be :

(Given : g = 10 ms^{-2})



(A) 10 N (B) 20 N (C) 25 N (D) 45 N

Q68. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

A block of mass 40 kg slides over a surface, when a mass of 4 kg is suspended through an inextensible massless string passing over frictionless pulley as shown below.

The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is. (Given g = 10 ms^{-2}.)



(A) 1 ms^{-2} (B) 1/5 ms^{-2} (C) 4/5 ms^{-2} (D) 8/11 ms^{-2}

Q69. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of 'a' will be

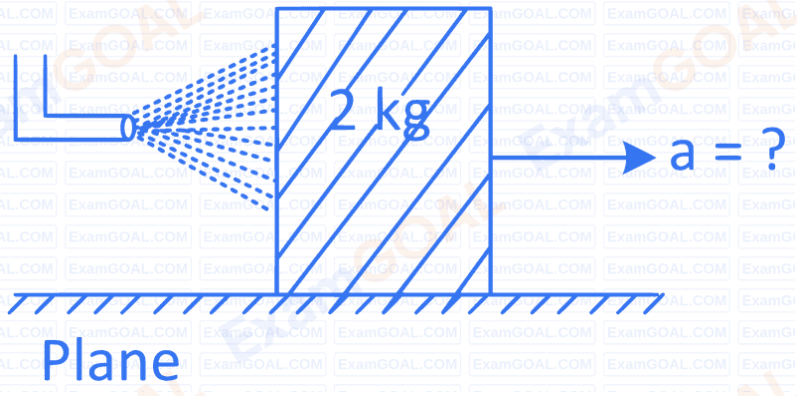
(A) $\frac{g}{4}$ (B) $\frac{g}{2}$ (C) $\frac{3g}{4}$ (D) g

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Q70. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

A block of metal weighing 2 kg is resting on a frictionless plane (as shown in figure). It is struck by a jet releasing water at a rate of 1 kgs^{-1} and at a speed of 10 ms^{-1}. Then, the initial acceleration of the block, in ms^{-2}, will be :

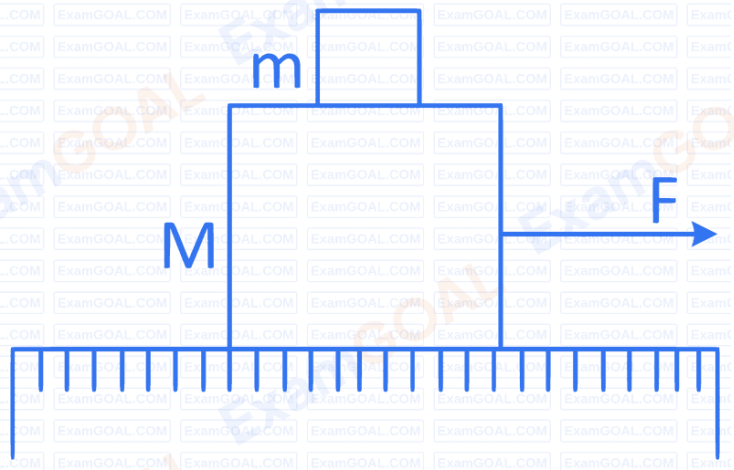


(A) 3 (B) 6 (C) 5 (D) 4

Q71. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

A system of two blocks of masses m = 2 kg and M = 8 kg is placed on a smooth table as shown in figure. The coefficient of static friction between two blocks is 0.5. The maximum horizontal force F that can be applied to the block of mass M so that the blocks move together will be :



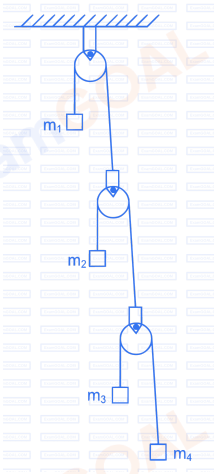
(A) 9.8 N (B) 39.2 N (C) 49 N (D) 78.4 N

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Q72. JEE Main 2022 (Online) 26th June Evening Shift

In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement?

Open solution



(A) $4a_1 + 2a_2 + a_3 + a_4 = 0$

(B) $a_1 + 4a_2 + 3a_3 + a_4 = 0$

(C) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$

(D) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

Q73. JEE Main 2022 (Online) 26th June Morning Shift

A person is standing in an elevator. In which situation, he experiences weight loss?

Open solution

(A) When the elevator moves upward with constant acceleration

(B) When the elevator moves downward with constant acceleration

(C) When the elevator moves upward with uniform velocity

(D) When the elevator moves downward with uniform velocity

Q74. JEE Main 2022 (Online) 24th June Evening Shift

An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to : [Use g = 10 ms^{-2}].

Open solution

(A) 1 : 1 (B) $\sqrt{2} : \sqrt{3}$ (C) $\sqrt{3} : \sqrt{2}$ (D) 2 : 3

Q75. JEE Main 2022 (Online) 24th June Morning Shift

A block of mass 10 kg starts sliding on a surface with an initial velocity of 9.8 ms^{-1}. The coefficient of friction between the surface and block is 0.5. The distance covered by the block before coming to rest is :

Open solution

[use g = 9.8 ms^{-2}]

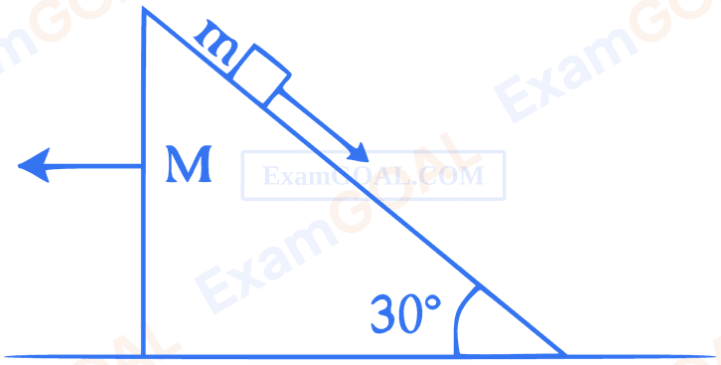
(A) 4.9 m (B) 9.8 m (C) 12.5 m (D) 19.6 m

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Q76. JEE Main 2021 (Online) 1st September Evening Shift

Open solution

A block of mass m slides on the wooden wedge, which in turn slides backward on the horizontal surface. The acceleration of the block with respect to the wedge is : Given m = 8 kg, M = 16 kg. Assume all the surfaces shown in the figure to be frictionless.



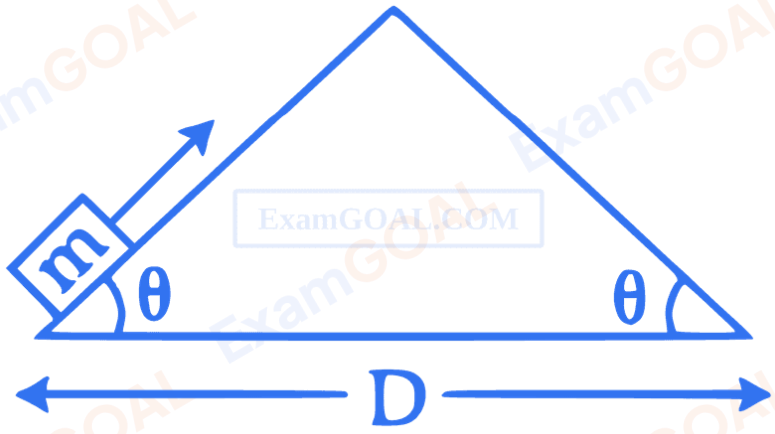
(A) $\frac{4}{3}g$ (B) $\frac{6}{5}g$ (C) $\frac{3}{5}g$ (D) $\frac{2}{3}g$

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Q77. JEE Main 2021 (Online) 1st September Evening Shift

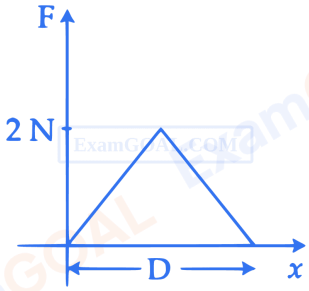
Open solution

An object of mass 'm' is being moved with a constant velocity under the action of an applied force of 2N along a frictionless surface with following surface profile.

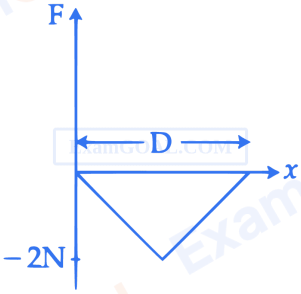


The correct applied force vs distance graph will be :

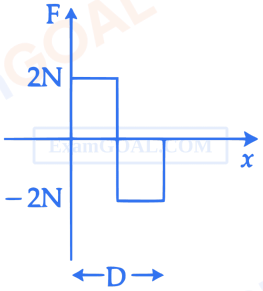
(A)



(B)

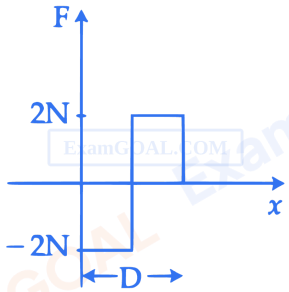


(C)



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(D)



Q78. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

Statement : 1

If three forces $\vec{F_1}, \vec{F_2}$ and $\vec{F_3}$ are represented by three sides of a triangle and $\vec{F_1} + \vec{F_2} = -\vec{F_3}$, then these three forces are concurrent forces and satisfy the condition for equilibrium.

Statement : 2

A triangle made up of three forces $\vec{F_1}, \vec{F_2}$ and $\vec{F_3}$ as its sides taken in the same order, satisfy the condition for translatory equilibrium.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement - I is false but Statement - II is true

(B) Statement - I is true but Statement - II is false

(C) Both Statement-I and Statement-II are false

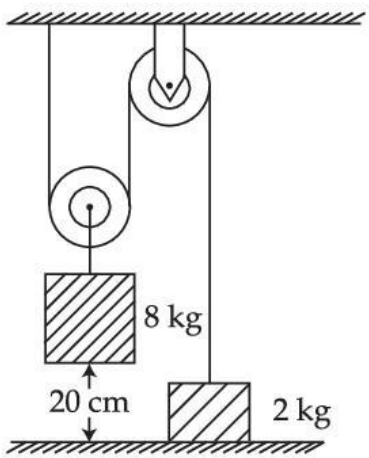
(D) Both Statement-I and Statement-II are true

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Q79. JEE Main 2021 (Online) 27th August Evening Shift

The boxes of masses 2 kg and 8 kg are connected by a massless string passing over smooth pulleys. Calculate the time taken by box of mass 8 kg to strike the ground starting from rest. (use g = 10 m/s^2)

Open solution



(A) 0.34 s (B) 0.2 s (C) 0.25 s (D) 0.4 s

Q80. JEE Main 2021 (Online) 27th July Evening Shift

A particle of mass M originally at rest is subjected to a force whose direction is constant but magnitude varies with time according to the relation

$$F = F_0\left[1-\left(\frac{t-T}{T}\right)^2\right]$$

Where F_0 and T are constants. The force acts only for the time interval 2T. The velocity v of the particle after time 2T is :

Open solution

(A) $2F_0T/M$ (B) $F_0T/2M$ (C) $4F_0T/3M$ (D) $F_0T/3M$

Q81. JEE Main 2021 (Online) 25th July Evening Shift

A force $\vec{F} = (40\hat{i} + 10\hat{j})N$ acts on a body of mass 5 kg. If the body starts from rest, its position vector $\vec{r}$ at time t = 10 s, will be :

Open solution

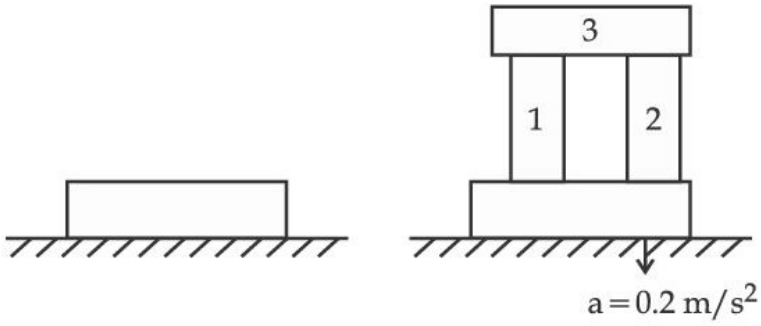
(A) $(100\hat{i} + 400\hat{j})m$ (B) $(100\hat{i} + 100\hat{j})m$ (C) $(400\hat{i} + 100\hat{j})m$ (D) $(400\hat{i} + 400\hat{j})m$

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Q82. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

A steel block of 10 kg rests on a horizontal floor as shown. When three iron cylinders are placed on it as shown, the block and cylinders go down with an acceleration 0.2 m/s^2. The normal reaction R' by the floor if mass of the iron cylinders are equal and of 20 kg each, is _____ N. [Take g = 10 m/s^2 and μ_s = 0.2]



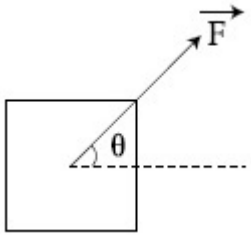
(A) 686 (B) 684 (C) 714 (D) 716

Q83. JEE Main 2021 (Online) 16th March Morning Shift

Open solution

A block of mass m slides along a floor while a force of magnitude F is applied to it at an angle θ as shown in figure. The coefficient of kinetic friction is μ_k. Then, the block's acceleration 'a' is given by :

(g is acceleration due to gravity)



(A) $\frac{F}{m}\cos\theta + \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

(B) $\frac{F}{m}\cos\theta - \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

(C) $-\frac{F}{m}\cos\theta - \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

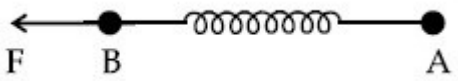
(D) $\frac{F}{m}\cos\theta - \mu_K\left(g + \frac{F}{m}\sin\theta\right)$

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Q84. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

Two masses A and B, each of mass M are fixed together by a massless spring. A force acts on the mass B as shown in the figure. If the mass A starts moving away from mass B with acceleration 'a', then the acceleration of mass B will be :



(A) $\frac{MF}{F+Ma}$ (B) $\frac{F+Ma}{M}$ (C) $\frac{Ma-F}{M}$ (D) $\frac{F-Ma}{M}$

Q85. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin i.e. ma = $-\alpha x^2$. The distance at which the particle stops :

(A) $\left[\frac{3mv_0^2}{2\alpha}\right]^{\frac{1}{3}}$ (B) $\left(\frac{2v_0}{3\alpha}\right)^{\frac{1}{3}}$

(C) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{2}}$ (D) $\left(\frac{2v_0^2}{3\alpha}\right)^{\frac{1}{2}}$

Q86. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

A particle moving in the xy plane experiences a velocity dependent force $\overrightarrow{F} = k\left(v_y\hat{i} + v_x\hat{j}\right)$, where v_x and v_y are the x and y components of its velocity $\overrightarrow{v}$. If $\overrightarrow{a}$ is the acceleration of the particle, then which of the following statements is true for the particle?

(A) kinetic energy of particle is constant in time

(B) quantity $\overrightarrow{v} \times \overrightarrow{a}$ is constant in time

(C) quantity $\overrightarrow{v}.\overrightarrow{a}$ is constant in time

(D) $\overrightarrow{F}$ arises due to a magnetic field

Q87. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

An insect is at the bottom of a hemispherical ditch of radius 1 m. It crawls up the ditch but starts slipping after it is at height h from the bottom. If the coefficient of friction between the ground and the insect is 0.75, then h is :
(g = 10 ms^{-2})

(A) 0.45 m (B) 0.60 m (C) 0.20 m (D) 0.80 m

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Q88. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

A small ball of mass m is thrown upward with velocity u from the ground. The ball experiences a resistive force mkv^2 where v is its speed. The maximum height attained by the ball is :

(A) $\frac{1}{k}\tan^{-1}\frac{ku^2}{2g}$ (B) $\frac{1}{2k}\tan^{-1}\frac{ku^2}{g}$

(C) $\frac{1}{2k}\ln\left(1+\frac{ku^2}{g}\right)$ (D) $\frac{1}{k}\ln\left(1+\frac{ku^2}{2g}\right)$

Q89. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

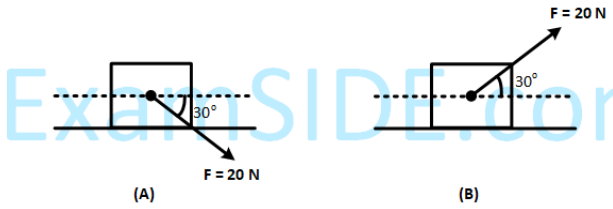
A mass of 10 kg is suspended by a rope of length 4 m, from the ceiling. A force F is applied horizontally at the mid point of the rope such that the top half of the rope makes an angle of 45° with the vertical. Then F equal : (Take g = 10 ms^{-2} and the rope to be massless)

(A) 100 N (B) 75 N (C) 90 N (D) 70 N

Q90. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

A block of mass 5 kg is (i) pushed in case (A) and (ii) pulled in case (B), by a force F = 20 N, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is μ = 0.2. The difference between the accelerations of the blocks, in case (B) and case (A) will be : (g = 10 ms^{-2})



(A) 3.2 ms^{-2} (B) 0.8 ms^{-2} (C) 0 ms^{-2} (D) 0.4 ms^{-2}

Q91. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

A spring whose unstretched length is l has a force constant k. The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constant, k_1 and k_2 will be :

(A) $\frac{1}{n^2}$ (B) $\frac{1}{n}$ (C) n^2 (D) n

Q92. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

A bullet of mass 20 g has an initial speed of 1 ms^{-1} , just before it starts penetrating a mud wall of thickness 20 cm. If the wall offers a mean resistance of 2.5×10^{-2} N, the speed of the bullet after emerging from the other side of the wall is close to :

(A) 0.3 ms^{-1} (B) 0.1 ms^{-1} (C) 0.7 ms^{-1} (D) 0.4 ms^{-1}

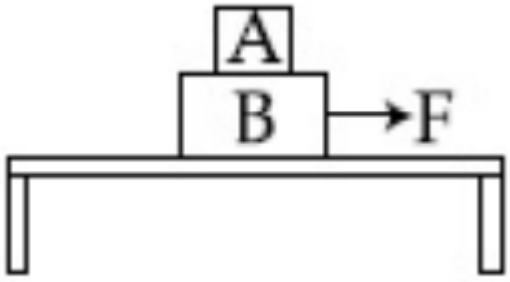
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Q93. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

Two blocks A and B of masses m_A = 1 kg and m_B = 3 kg are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2. The maximum force F that can be applied on B horizontally, so that the block A does not slide over the block B is :

[Take g = 10 m/s^2]



(A) 8 N (B) 16 N (C) 40 N (D) 12 N

Q94. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

A ball is thrown upward with an initial velocity V_0 from the surface of the earth. The motion of the ball is affected by a drag force equal to $m\gamma u^2$ (where m is mass of the ball, u is its instantaneous velocity and γ is a constant). Time taken by the ball to rise to its zenith is :

(A) $\frac{1}{\sqrt{\gamma g}}\tan^{-1}\left(\sqrt{\frac{\gamma}{g}}V_0\right)$

(B) $\frac{1}{\sqrt{\gamma g}}ln\left(1+\sqrt{\frac{\gamma}{g}}V_0\right)$

(C) $\frac{1}{\sqrt{\gamma g}}\sin^{-1}\left(\sqrt{\frac{\gamma}{g}}V_0\right)$

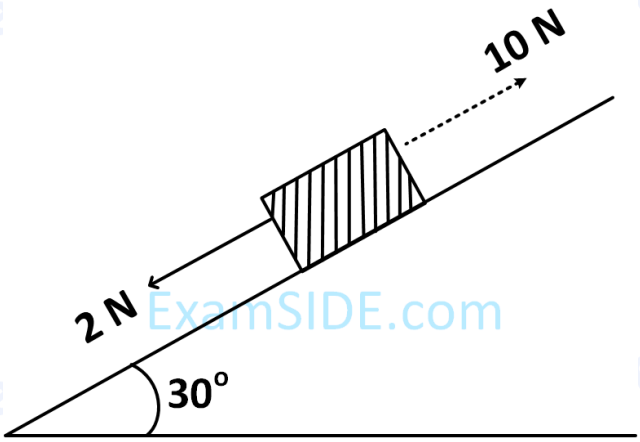
(D) $\frac{1}{\sqrt{2\gamma g}}\tan^{-1}\left(\sqrt{\frac{2\gamma}{g}}V_0\right)$

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Q95. JEE Main 2019 (Online) 12th January Evening Slot

A block kept on a rough inclined plane, as shown in the figure, remains at rest upto a maximum force 2 N down the inclined plane. The maximum external force up the inclined plane that does not move the block is 10 N. The coefficient of static friction between the block and the plane is : [Take g = 10 m/s^2]

Open solution



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(A) $\frac{\sqrt{3}}{4}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{2}{3}$

Q96. JEE Main 2019 (Online) 9th January Evening Slot

A mass of 10 kg is suspended vertically by a rope from the roof. When a horizontal force is applied on the rope at some point, the rope deviates at an angle of 45° at the roof point. If the suspended mass is at equilibrium, the magnitude of the force applied is (g = 10 ms^{-2})

Open solution

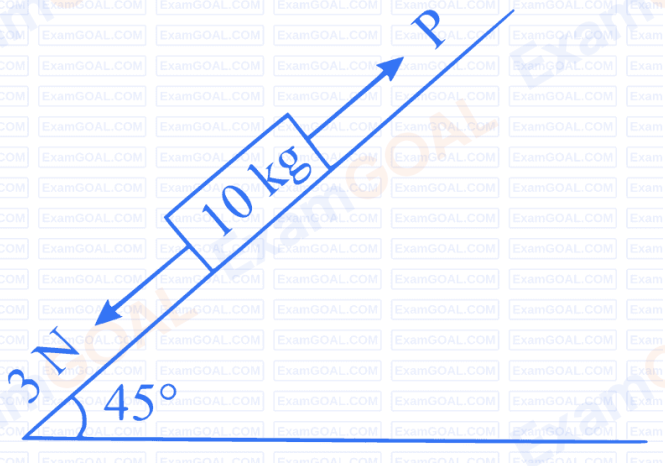
(A) 200 N (B) 140 N (C) 70 N (D) 100 N

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Q97. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

A block of mass 10 kg is kept on a rough inclined plane as shown in the figure. A force of 3 N is applied on the block. The coefficient of static friction between the plane and the block is 0.6. What should be the minimum value of force P, such that the block does not move downward? (take g = 10 ms^{-2})

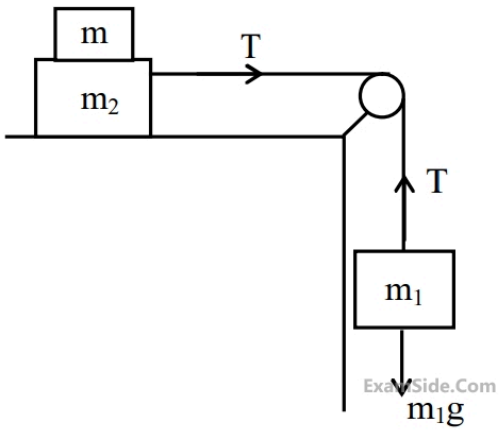


(A) 32 N (B) 18 N (C) 23 N (D) 25 N

Q98. JEE Main 2018 (Offline)

Open solution

Two masses m_1 = 5 kg and m_2 = 10 kg, connected by an inextensible string over a frictionless pulley, are moving as shown in the figure. The coefficient of friction of horizontal surface is 0.15. The minimum weight m that should be put on top of m_2 to stop the motion is :



(A) 10.3 kg (B) 18.3 kg (C) 27.3 kg (D) 43.3 kg

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Q99. JEE Main 2018 (Online) 15th April Evening Slot

A body of mass 2 kg slides down with an acceleration of 3 m/s^2 on a rough inclined plane having a slope of 30°. The external force required to take the same body up the plane with the same acceleration will be : (g = 10 m/s^2)

Open solution

(A) 14 N (B) 20 N (C) 6 N (D) 4 N

Q100. JEE Main 2018 (Online) 15th April Morning Slot

A given object takes n times more time to slide down a 45° rough inclined plane as it takes to slide down a perfectly smooth 45° incline. The coefficient of kinetic friction between the object and the incline is :

Open solution

(A) $\frac{1}{2-n^2}$ (B) $1-\frac{1}{n^2}$ (C) $\sqrt{1-\frac{1}{n^2}}$ (D) $\sqrt{\frac{1}{1-n^2}}$

Q101. JEE Main 2017 (Offline)

A body of mass m = 10^{-2} kg is moving in a medium and experiences a frictional force F = $-kv^2$. Its initial speed is v_0 = 10 ms^{-1}. If, after 10 s, its energy is $\frac{1}{8}mv_0^2$, the value of k will be:

Open solution

(A) 10^{-1} kg m^{-1} s^{-1} (B) 10^{-3} kg m^{-1} (C) 10^{-3} kg s^{-1} (D) 10^{-4} kg m^{-1}

Q102. JEE Main 2016 (Online) 10th April Morning Slot

A particle of mass m is acted upon by a force F given by the empirical law F = $\frac{R}{t^2}v(t)$. If this law is to be tested experimentally by observing the motion starting from rest, the best way is to plot :

Open solution

(A) v(t) against t^2 (B) log v(t) against $\frac{1}{t^2}$

(C) log v(t) against t (D) log v(t) against $\frac{1}{t}$

Q103. JEE Main 2016 (Online) 9th April Morning Slot

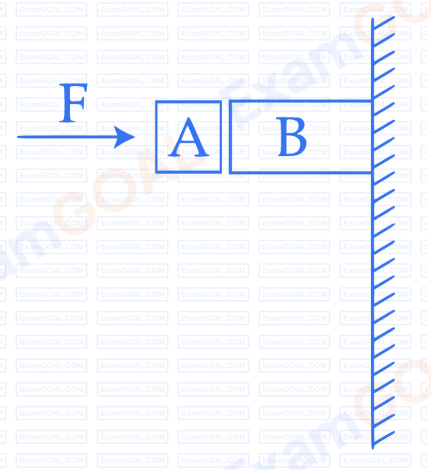
A rocket is fired vertically from the earth with an acceleration of 2g, where g is the gravitational acceleration. On an inclined plane inside the rocket, making an angle θ with the horizontal, a point object of mass m is kept. The minimum coefficient of friction μ_{min} between the mass and the inclined surface such that the mass does not move is :

Open solution

(A) tanθ (B) 2tanθ (C) 3tanθ (D) tan2θ

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Q104. JEE Main 2015 (Offline)



Open solution

Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force applied by the wall on block B is :

(A) $120\ N$ (B) $150\ N$ (C) $100\ N$ (D) $80\ N$

Q105. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

A heavy box is to be dragged along a rough horizontal floor. To do so, person A pushes it at an angle 30° from the horizontal and requires a minimum force F_A, while person B pulls the box at an angle 60° from the horizontal and needs minimum force F_B. If the coefficient of friction between the box and the floor is $\frac{\sqrt{3}}{5}$, the ratio $\frac{F_A}{F_B}$ is :

(A) $\sqrt{3}$ (B) $\frac{5}{\sqrt{3}}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{2}{\sqrt{3}}$

Q106. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

A Body of mass 5 kg under the action of constant force $\vec{F} = F_x\,\hat{i} + F_y\,\hat{j}$ has velocity at t = 0 s as $\vec{v} = (6\hat{i} - 2\hat{j})$ m/s and at t = 10 s as $\vec{v} = +6\hat{j}$ m/s. The force $\vec{F}$ is :

(A) $(-3\hat{i} + 4\hat{j})$ N (B) $\left(-\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}\right)N$

(C) $(3\hat{i} - 4\hat{j})$ N (D) $\left(\frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}\right)N$

Q107. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Water is flowing at a speed of 1.5 ms^{-1} through a horizontal tube of cross-sectional area 10^{-2} m^2 and you are trying to stop the flow by your palm. Assuming that the water stops immediately after hitting the palm, the minimum force that you must exert should be (density of water = 10^3 kgm^{-3}).

(A) 15 N (B) 22.5 N (C) 33.7 N (D) 45 N

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Q108. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

A block A of mass 4 kg is placed on another block B of mass 5 kg, and the block B rests on a smooth horizontal table. If the minimum force that can be applied on A so that both the blocks move together is 12 N, the maximum force that can be applied on B for the blocks to move together will be :

(A) 30 N (B) 25 N (C) 27 N (D) 48 N

Q109. JEE Main 2014 (Offline)

Open solution

A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is:

(A) $\frac{1}{6}m$ (B) $\frac{2}{3}m$ (C) $\frac{1}{3}m$ (D) $\frac{1}{2}m$

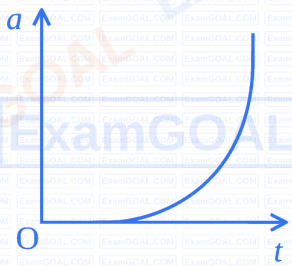
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Q110. JEE Main 2013 (Online) 25th April Morning Slot

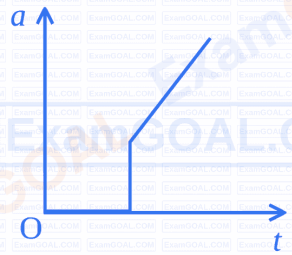
Open solution

A block is placed on a rough horizontal plane. A time dependent horizontal force $F = kt$ acts on the block, where k is a positive constant. The acceleration - time graph of the block is :

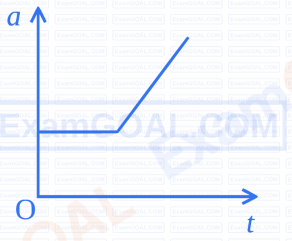
(A)



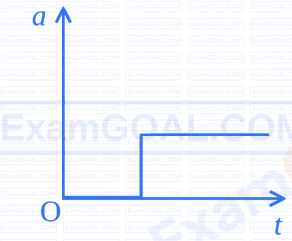
(B)



(C)



(D)

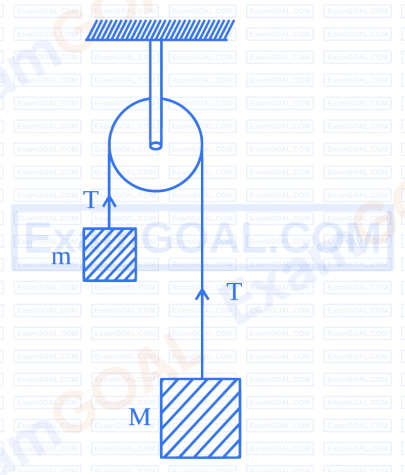


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Q111. JEE Main 2013 (Online) 25th April Morning Slot

Two blocks of masses m and M are connected by means of a metal wire of cross-sectional area A passing over a frictionless fixed pulley as shown in the figure. The system is then released. If M = 2m, then the stress produced in the wire is:

Open solution

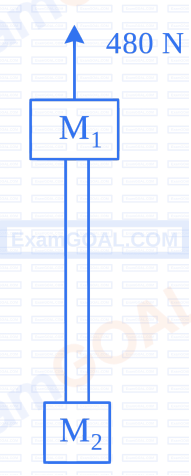


(A) $\frac{2mg}{3A}$ (B) $\frac{4mg}{3A}$ (C) $\frac{mg}{A}$ (D) $\frac{3mg}{4A}$

Q112. JEE Main 2013 (Online) 22th April Morning Slot

Two blocks of mass $M_1 = 20$ kg and $M_2 = 12$ kg are connected by a metal rod of mass 8 kg. The system is pulled vertically up by applying a force of 480 N as shown. The tension at the mid-point of the rod is:

Open solution



(A) 144 N (B) 96 N (C) 240 N (D) 192 N

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Q113. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

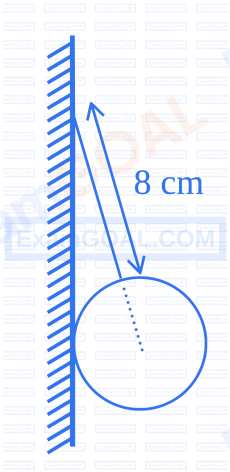
A body starts from rest on a long inclined plane of slope 45°. The coefficient of friction between the body and the plane varies as $\mu = 0.3x$, where x is distance travelled down the plane. The body will have maximum speed (for $g = 10$ m/s^2) when x =

(A) 9.8 m (B) 27 m (C) 12 m (D) 3.33 m

Q114. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

A uniform sphere of weight W and radius 5 cm is being held by a string as shown in the figure. The tension in the string will be :



(A) $12\frac{W}{5}$ (B) $5\frac{W}{12}$ (C) $13\frac{W}{5}$ (D) $13\frac{W}{12}$

Q115. AIEEE 2012 (Online) 26th May

Open solution

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: If you push on a cart being pulled by a horse so that it does not move, the cart pushes you back with an equal and opposite force.
Statement 2: The cart does not move because the force described in statement 1 cancel each other.

(A) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

(B) Statement 1 is false, Statement 2 is true.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

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Q116. AIEEE 2012 (Online) 19th May

A block of weight W rests on a horizontal floor with coefficient of static friction μ. It is desired to make the block move by applying minimum amount of force. The angle θ from the horizontal at which the force should be applied and magnitude of the force F are respectively.

Open solution

(A) $\theta = \tan^{-1}(\mu),\ F = \frac{\mu W}{\sqrt{1+\mu^2}}$

(B) $\theta = \tan^{-1}\left(\frac{1}{\mu}\right),\ F = \frac{\mu W}{\sqrt{1+\mu^2}}$

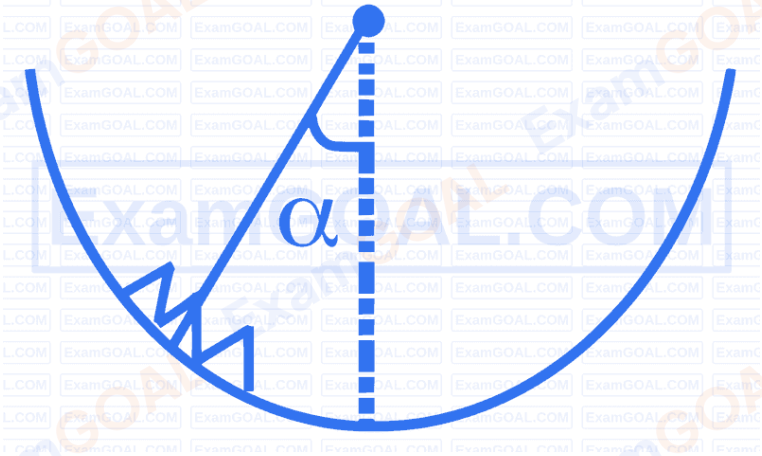
(C) $\theta = 0,\ F = \mu W$

(D) $\theta = \tan^{-1}\left(\frac{\mu}{1+\mu}\right),\ F = \frac{\mu W}{1+\mu}$

Q117. AIEEE 2012 (Online) 12th May

An insect crawls up a hemispherical surface very slowly. The coefficient of friction between the insect and the surface is $1/3$. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α so that the insect does not slip is given by

Open solution



(A) $\cot\alpha = 3$

(B) $\sec\alpha = 3$

(C) $\operatorname{cosec}\alpha = 3$

(D) $\cos\alpha = 3$

Q118. AIEEE 2012 (Online) 7th May

A car of mass $1000\ \text{kg}$ is moving at a speed of $30\ \text{m/s}$. Brakes are applied to bring the car to rest. If the net retarding force is $5000\ \text{N}$, the car comes to stop after travelling d m in t s. Then

Open solution

(A) $d = 150,\ t = 5$

(B) $d = 120,\ t = 8$

(C) $d = 180,\ t = 6$

(D) $d = 90,\ t = 6$

Q119. AIEEE 2012 (Online) 7th May

An engine pumps water continuously through a hose. Water leaves the hose with velocity v and m is mass per unit length of the water jet. If this jet hits a surface and came to rest instantaneously, the force on the surface is

Open solution

(A) mv^3

(B) mv^2

(C) $\frac{1}{2}mv^2$

(D) $\frac{1}{2}mv^3$

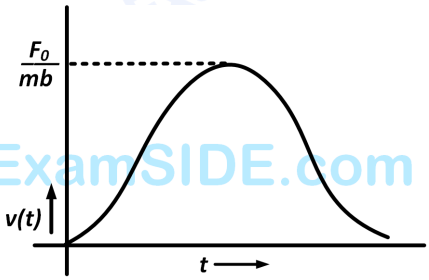
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Q120. AIEEE 2012

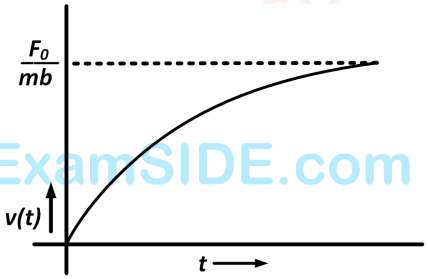
Open solution

A particle of mass m is at rest at the origin at time $t = 0$. It is subjected to a force $F(t) = F_0 e^{-bt}$ in the x direction. Its speed $v(t)$ is depicted by which of the following curves?

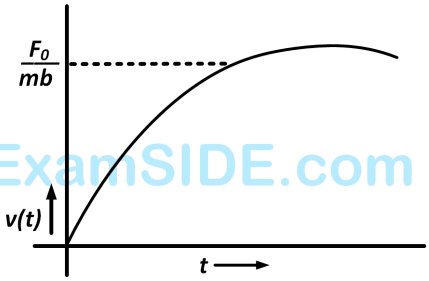
(A)



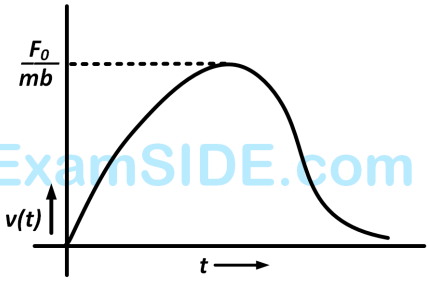
(B)



(C)



(D)

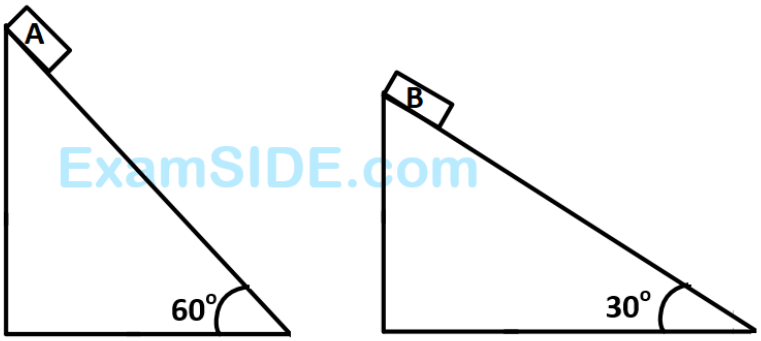


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Q121. AIEEE 2010

Two fixed frictionless inclined planes making an angle 30° and 60° with the vertical are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B ?

Open solution



ExamSIDE.com

(A) $4.9ms^{-2}$ in horizontal direction

(B) $9.8ms^{-2}$ in vertical direction

(C) Zero

(D) $4.9ms^{-2}$ in vertical direction

Q122. AIEEE 2007

A block of mass m is connected to another block of mass M by a spring (massless) of spring constant k. The blocks are kept on a smooth horizontal plane. Initially the blocks are at rest and the spring is unstretched. Then a constant force F starts acting on the block of mass M to pull it. Find the force on the block of mass m.

Open solution

(A) $\frac{MF}{(m+M)}$

(B) $\frac{mF}{M}$

(C) $\frac{(M+m)F}{m}$

(D) $\frac{mF}{(m+M)}$

Q123. AIEEE 2005

A smooth block is released at rest on a 45° incline and then slides a distance $'d'$. The time taken to slide is $'n'$ times as much to slide on rough incline than on a smooth incline. The coefficient of friction is

Open solution

(A) $\mu_k = \sqrt{1 - \frac{1}{n^2}}$

(B) $\mu_k = 1 - \frac{1}{n^2}$

(C) $\mu_s = \sqrt{1 - \frac{1}{n^2}}$

(D) $\mu_s = 1 - \frac{1}{n^2}$

Q124. AIEEE 2005

A particle of mass 0.3 kg subjected to a force $F = -kx$ with $k = 15\ N/m$. What will be its initial acceleration if it is released from a point 20 cm away from the origin?

Open solution

(A) $15\ \ m/s^2$

(B) $3\ \ m/s^2$

(C) $10\ \ m/s^2$

(D) $5\ \ m/s^2$

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Q125. AIEEE 2005

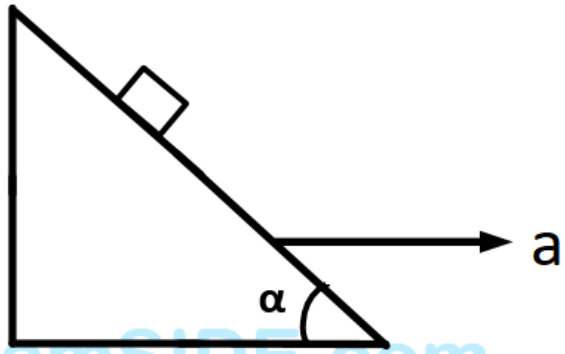
Consider a car moving on a straight road with a speed of $100\ m/s$. The distance at which car can be stopped is $[\mu_k = 0.5]$

(A) $1000\ m$ (B) $800\ m$ (C) $400\ m$ (D) $100\ m$

Open solution

Q126. AIEEE 2005

A block is kept on a frictionless inclined surface with angle of inclination $'\alpha'$. The incline is given an acceleration a to keep the block stationary. Then a is equal to



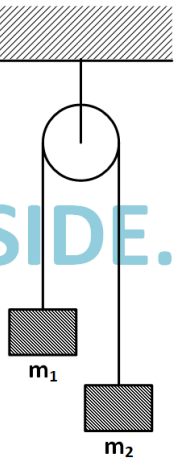
ExamSIDE.com

(A) $g\ cosec\ \alpha$ (B) $g/tan\ \alpha$ (C) $g\ tan\ \alpha$ (D) g

Open solution

Q127. AIEEE 2004

Two masses $m_1 = 5kg$ and $m_2 = 4.8kg$ tied to a string are hanging over a light frictionless pulley. What is the acceleration of the masses when left free to move? $(g = 9.8m/s^2)$



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(A) $5\ m/s^2$ (B) $9.8\ m/s^2$ (C) $0.2\ m/s^2$ (D) $4.8\ m/s^2$

Open solution

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Q128. AIEEE 2004

A block rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If the frictional force on the block is $10\ N$, the mass of the block (in kg) is $(take\ \ g = 10\ m/s^2)$

(A) 1.6 (B) 4.0 (C) 2.0 (D) 2.5

Open solution

Q129. AIEEE 2003

A spring balance is attached to the ceiling of a lift. A man hangs his bag on the spring and the spring reads $49\ N$, when the lift is stationary. If the lift moves downward with an acceleration of $5m/s^2$, the reading of the spring balance will be

(A) $24\ N$ (B) $74\ N$ (C) $15\ N$ (D) $49\ N$

Open solution

Q130. AIEEE 2003

A marble block of mass $2\ kg$ lying on ice when given a velocity of $6\ m/s$ is stopped by friction in $10\ s$. Then the coefficient of friction is

(A) 0.02 (B) 0.03 (C) 0.04 (D) 0.06

Open solution

Q131. AIEEE 2003

A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m. If a force P is applied at the free end of the rope, the force exerted by the rope on the block is

(A) $\frac{Pm}{M+m}$ (B) $\frac{Pm}{M-m}$ (C) P (D) $\frac{PM}{M+m}$

Open solution

Q132. AIEEE 2003

A light spring balance hangs from the hook of the other light spring balance and a block of mass M kg hangs from the former one. Then the true statement about the scale reading is

(A) Both the scales read $M\ kg$ each

(B) The scale of the lower one reads $M\ kg$ and of the upper one zero

(C) The reading of the two scales can be anything but the sum of the reading will be $M\ kg$

(D) Both the scales read $M/2\ kg$ each

Open solution

Q133. AIEEE 2003

A rocket with a lift-off mass $3.5 \times 10^4\ kg$ is blasted upwards with an initial acceleration of $10m/s^2$. Then the initial thrust of the blast is

(A) $3.5 \times 10^5 N$ (B) $7.0 \times 10^5 N$ (C) $14.0 \times 10^5 N$ (D) $1.75 \times 10^5 N$

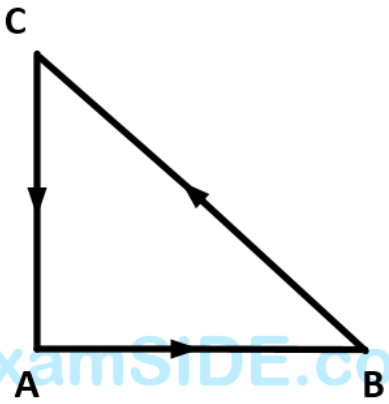
Open solution

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Q134. AIEEE 2003

Three forces start acting simultaneously on a particle moving with velocity, $\overrightarrow{v}$. These forces are represented in magnitude and direction by the three sides of a triangle ABC. The particle will now move with velocity

Open solution



(A) less than $\overrightarrow{v}$

(B) greater than $\overrightarrow{v}$

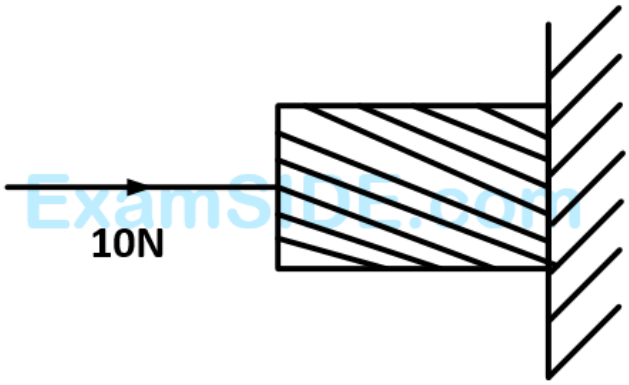
(C) $|v|$ in the direction of the largest force BC

(D) $\overrightarrow{v}$, remaining unchanged

Q135. AIEEE 2003

A horizontal force of $10\ N$ is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is

Open solution



(A) $20N$ (B) $50N$ (C) $100N$ (D) $2N$

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Q136. AIEEE 2002

A lift is moving down with acceleration a. A man in the lift drops a ball inside the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground are respectively

Open solution

(A) g, g (B) $g-a, g-a$ (C) $g-a, g$ (D) a, g

Q137. AIEEE 2002

When forces $F_1,\ F_2,\ F_3$ are acting on a particle of mass m such that F_2 and F_3 are mutually perpendicular, then the particle remains stationary. If the force F_1 is now removed then the acceleration of the particle is

Open solution

(A) F_1/m (B) F_2F_3/mF_1 (C) $(F_2-F_3)/m$ (D) F_2/m

Q138. AIEEE 2002

A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 (vertically). If the acceleration of the system is $g/8$, then the ratio of the masses is

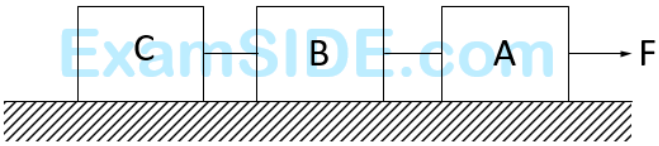
Open solution

(A) $8:1$ (B) $9:7$ (C) $4:3$ (D) $5:3$

Q139. AIEEE 2002

Three identical blocks of masses $m = 2\ kg$ are drawn by a force $F = 10.2\ N$ with an acceleration of $0.6\ ms^{-2}$ on a frictionless surface, then what is the tension (in N) in the string between the blocks B and C?

Open solution



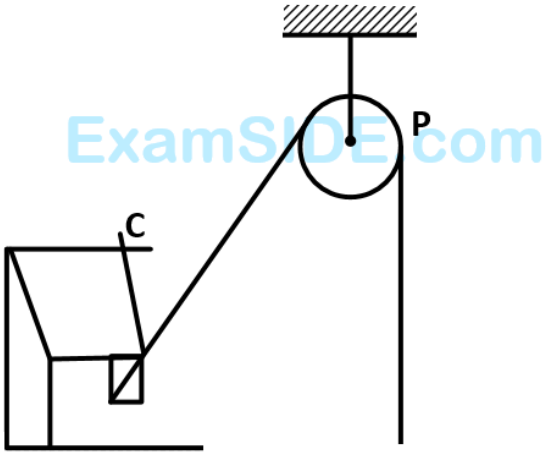
(A) 9.2 (B) 9.8 (C) 4 (D) 7.8

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Q140. AIEEE 2002

One end of a mass-less rope, which passes over a mass-less and friction-less pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is $360\ N$. With what value of maximum safe acceleration (in ms^{-2}) can a man of $60\ kg$ climb on the rope?

Open solution



(A) 16 (B) 6 (C) 4 (D) 8

Numerical 25 questions

Q141. JEE Main 2026 (Online) 6th April Evening Shift

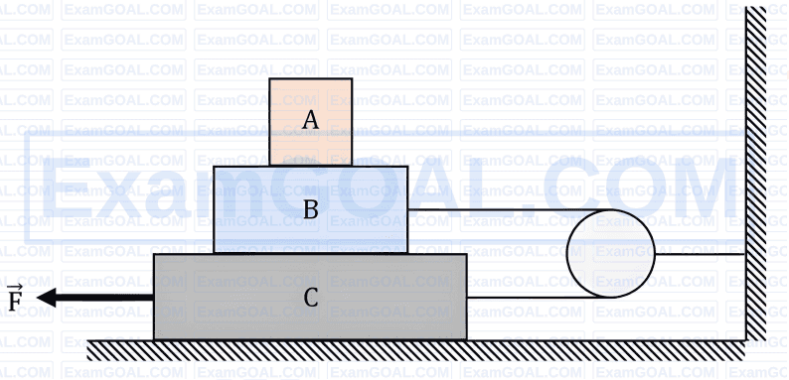
A block takes t time to slide down a plane inclined at 45° to the horizontal. If the surface is made smooth (frictionless), the block takes time $\frac{t}{2}$ to slide down the plane. The coefficient of friction between the block and the inclined plane is $\left(\frac{\alpha}{100}\right)$. The value of α is ____ .

Open solution

Q142. JEE Main 2026 (Online) 24th January Morning Shift

In the given figure the blocks A, B and C weigh $4\ \text{kg}, 6\ \text{kg}$ and 8 kg respectively. The co-efficient of sliding friction between any two surfaces is 0.5 . The force $\vec{F}$ required to slide the block C with constant speed is ____ N . (Use $g = 10\ \text{m/s}^2$)

Open solution

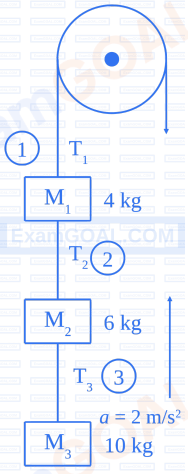


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Q143. JEE Main 2024 (Online) 5th April Morning Shift

Three blocks M_1, M_2, M_3 having masses $4\,kg, 6\,kg$ and $10\,kg$ respectively are hanging from a smooth pulley using rope 1, 2 and 3 as shown in figure. The tension in the rope $1, T_1$ when they are moving upward with acceleration of $2\,ms^{-2}$ is _____ N (if $g = 10\,m/s^2$).

Open solution

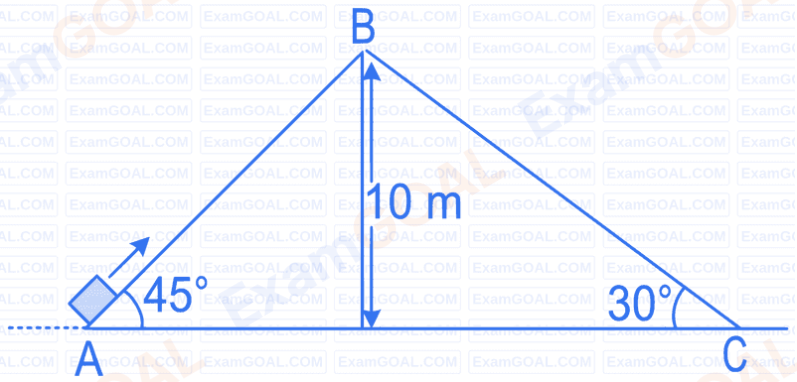


Q144. JEE Main 2022 (Online) 27th July Evening Shift

Two inclined planes are placed as shown in figure. A block is projected from the Point A of inclined plane AB along its surface with a velocity just sufficient to carry it to the top Point B at a height 10 m. After reaching the Point B the block slides down on inclined plane BC. Time it takes to reach to the point C from point A is $t(\sqrt{2}+1)$ s. The value of t is _____.

(use $g = 10\,m/s^2$)

Open solution

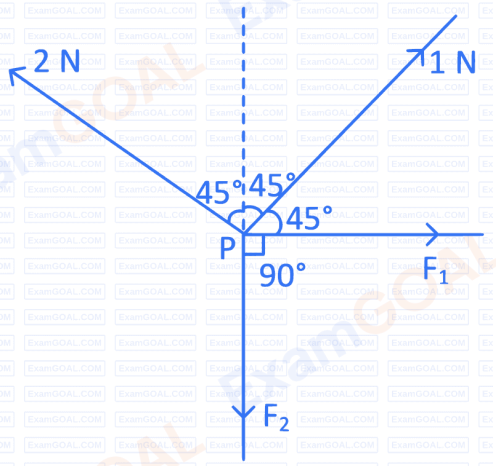


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Q145. JEE Main 2022 (Online) 25th July Morning Shift

Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force F_1 to F_2 is $1 : x$ where $x =$ _____.

Open solution

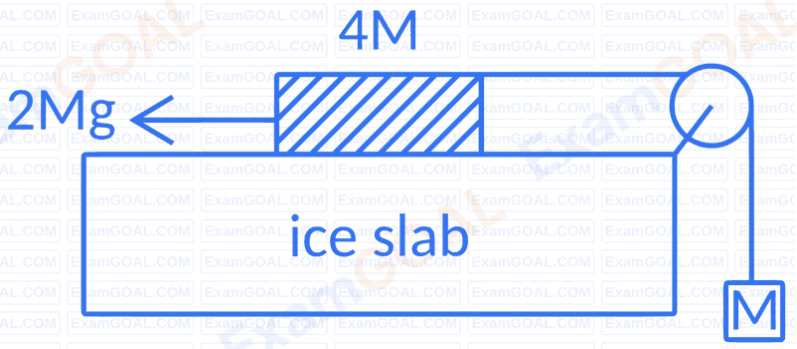


Q146. JEE Main 2022 (Online) 28th June Morning Shift

A hanging mass M is connected to a four times bigger mass by using a string-pulley arrangement, as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by 2 Mg force. In this situation, tension in the string is $\frac{x}{5}$ Mg for x = _____. Neglect mass of the string and friction of the block (bigger mass) with ice slab.

(Given g = acceleration due to gravity)

Open solution



Q147. JEE Main 2022 (Online) 27th June Evening Shift

A mass of 10 kg is suspended vertically by a rope of length 5 m from the roof. A force of 30 N is applied at the middle point of rope in horizontal direction. The angle made by upper half of the rope with vertical is $\theta = \tan^{-1}(x \times 10^{-1})$. The value of x is _____.

(Given, g = 10 m/s^2)

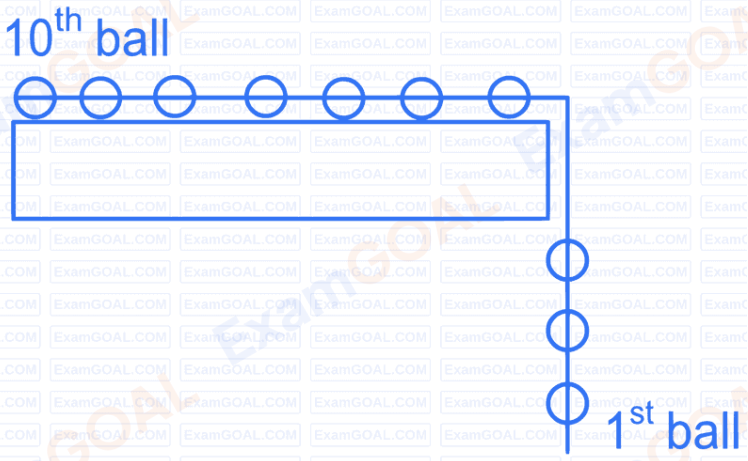
Open solution

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Q148. JEE Main 2022 (Online) 26th June Evening Shift

A system of 10 balls each of mass 2 kg are connected via massless and unstretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the 7th and 8th ball is _____ N when 6th ball just leaves the table.

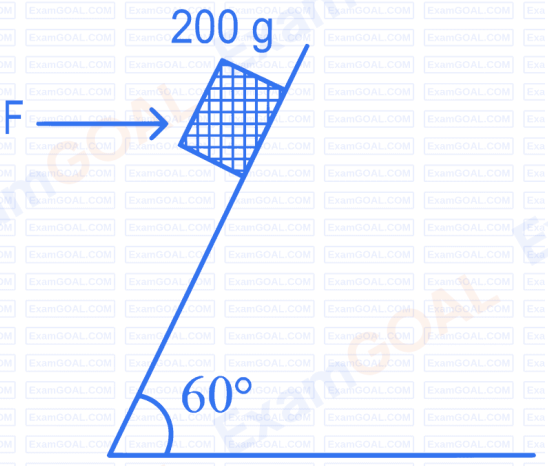
Open solution



Q149. JEE Main 2022 (Online) 25th June Evening Shift

A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force F = $\sqrt{x}$N as shown in figure.

Open solution



The value of x = _____.

Q150. JEE Main 2022 (Online) 25th June Morning Shift

A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is _____ m.

Open solution

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Q151. JEE Main 2022 (Online) 25th June Morning Shift

A force on an object of mass 100 g is $\left(10\hat{i}+5\hat{j}\right)$ N. The position of that object at t = 2 s is $\left(a\hat{i}+b\hat{j}\right)$ m after starting from rest. The value of $\frac{a}{b}$ will be _____.

Open solution

Q152. JEE Main 2021 (Online) 1st September Evening Shift

When a body slides down from rest along a smooth inclined plane making an angle of 30° with the horizontal, it takes time T. When the same body slides down from the rest along a rough inclined plane making the same angle and through the same distance, it takes time αT, where α is a constant greater than 1. The co-efficient of friction between the body and the rough plane is $\frac{1}{\sqrt{x}}\left(\frac{\alpha^2-1}{\alpha^2}\right)$ where x = _____.

Open solution

Q153. JEE Main 2021 (Online) 31st August Morning Shift

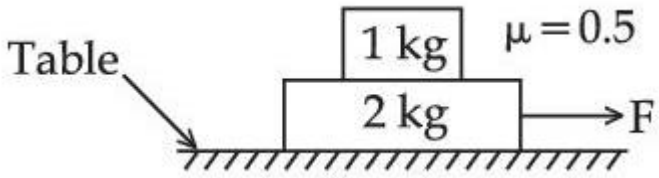
A car is moving on a plane inclined at 30° to the horizontal with an acceleration of 10 ms^{-2} parallel to the plane upward. A bob is suspended by a string from the roof of the car. The angle in degrees which the string makes with the vertical is _____. (Take g = 10 ms^{-2})

Open solution

Q154. JEE Main 2021 (Online) 26th August Evening Shift

The coefficient of static friction between two blocks is 0.5 and the table is smooth. The maximum horizontal force that can be applied to move the blocks together is ..... N. (take g = 10 ms^{-2})

Open solution



Q155. JEE Main 2021 (Online) 20th July Evening Shift

A body of mass 'm' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent. The value of x is _____.

Open solution

Q156. JEE Main 2021 (Online) 18th March Morning Shift

A bullet of mass 0.1 kg is fired on a wooden block to pierce through it, but it stops after moving a distance of 50 cm into it. If the velocity of bullet before hitting the wood is 10 m/s and it slows down with uniform deceleration, then the magnitude of effective retarding force on the bullet is 'x' N. The value of 'x' to the nearest integer is _____.

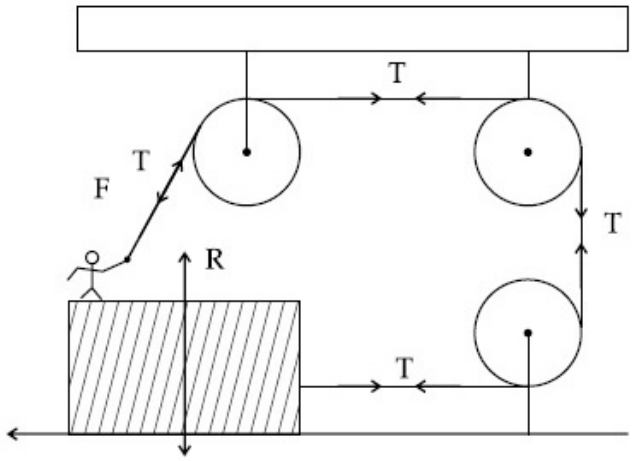
Open solution

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Q157. JEE Main 2021 (Online) 17th March Evening Shift

A boy of mass 4 kg is standing on a piece of wood having mass 5 kg. If the coefficient of friction between the wood and the floor is 0.5, the maximum force that the boy can exert on the rope so that the piece of wood does not move from its place is _____ N. (Round off to the Nearest Integer) [Take g = 10 ms^{-2}]

Open solution



Q158. JEE Main 2021 (Online) 17th March Evening Shift

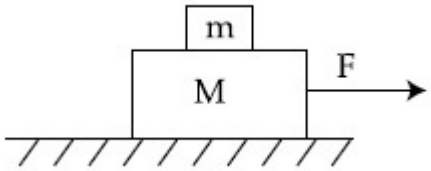
A body of mass 1 kg rests on a horizontal floor with which it has a coefficient of static friction $\frac{1}{\sqrt{3}}$. It is desired to make the body move by applying the minimum possible force F N. The value of F will be _____. (Round off to the Nearest Integer) [Take g = 10 ms^{-2}]

Open solution

Q159. JEE Main 2021 (Online) 17th March Morning Shift

Two blocks (m = 0.5 kg and M = 4.5 kg) are arranged on a horizontal frictionless table as shown in figure. The coefficient of static friction between the two blocks is $\frac{3}{7}$. Then the maximum horizontal force that can be applied on the larger block so that the blocks move together is _____ N. (Round off to the Nearest Integer) [Take g as 9.8 ms^{-2}]

Open solution



Q160. JEE Main 2021 (Online) 16th March Evening Shift

A body of mass 2 kg moves under a force of $\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$N. It starts from rest and was at the origin initially. After 4s, its new coordinates are (8, b, 20). The value of b is _____. (Round off to the Nearest Integer)

Open solution

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Q161. JEE Main 2021 (Online) 26th February Morning Shift

A person standing on a spring balance inside a stationary lift measures 60 kg. The weight of that person if the lift descends with uniform downward acceleration of 1.8 m/s^2 will be _____ N. [g = 10 m/s^2]

Open solution

Q162. JEE Main 2021 (Online) 26th February Morning Shift

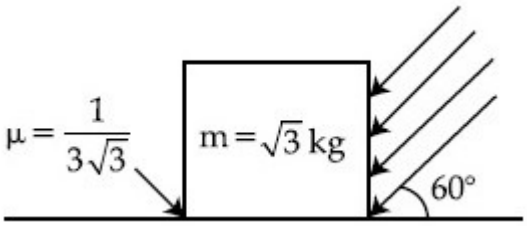
A boy pushes a box of mass 2 kg with a force $\vec{F} = \left(20\hat{i} + 10\hat{j}\right) N$ on a frictionless surface. If the box was initially at rest, then _____ m is displacement along the x-axis after 10s.

Open solution

Q163. JEE Main 2021 (Online) 26th February Morning Shift

As shown in the figure, a block of mass $\sqrt{3}$ kg is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move, will be 3x. The value of x will be _____.

[g = 10 m/s^2; sin60° = $\frac{\sqrt{3}}{2}$; cos60° = $\frac{1}{2}$]



Open solution

Q164. JEE Main 2021 (Online) 24th February Morning Shift

An inclined plane is bent in such a way that the vertical cross-section is given by $y = \frac{x^2}{4}$ where y is in vertical and x in horizontal direction. If the upper surface of this curved plane is rough with coefficient of friction μ = 0.5, the maximum height in cm at which a stationary block will not slip downward is _____ cm.

Open solution

Q165. JEE Main 2021 (Online) 24th February Morning Shift

The coefficient of static friction between a wooden block of mass 0.5 kg and a vertical rough wall is 0.2. The magnitude of horizontal force that should be applied on the block to keep it adhere to the wall will be _____ N. [g = 10 ms^{-2}]

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **B**	3. **A**	4. **A**	5. **D**
6. **C**	7. **B**	8. **B**	9. **D**	10. **B**
11. **C**	12. **C**	13. **D**	14. **B**	15. **D**
16. **D**	17. **A**	18. **B**	19. **C**	20. **B**
21. **C**	22. **A**	23. **C**	24. **A**	25. **C**
26. **A**	27. **B**	28. **B**	29. **A**	30. **D**
31. **A**	32. **A**	33. **D**	34. **A**	35. **B**
36. **A**	37. **D**	38. **A**	39. **D**	40. **A**
41. **B**	42. **C**	43. **C**	44. **D**	45. **A**
46. **D**	47. **D**	48. **A**	49. **D**	50. **C**
51. **A**	52. **C**	53. **B**	54. **D**	55. **D**
56. **A**	57. **B**	58. **D**	59. **B**	60. **A**
61. **C**	62. **B**	63. **B**	64. **A**	65. **C**
66. **A**	67. **D**	68. **D**	69. **C**	70. **C**
71. **C**	72. **A**	73. **B**	74. **B**	75. **B**
76. **D**	77. **C**	78. **D**	79. **D**	80. **C**
81. **C**	82. **A**	83. **B**	84. **D**	85. **A**
86. **B**	87. **C**	88. **C**	89. **A**	90. **B**
91. **B**	92. **C**	93. **B**	94. **A**	95. **C**
96. **D**	97. **A**	98. **C**	99. **B**	100. **B**
101. **D**	102. **D**	103. **A**	104. **A**	105. **D**
106. **A**	107. **B**	108. *Scan QR*	109. **A**	110. **B**
111. **B**	112. **D**	113. **D**	114. **D**	115. **C**
116. **A**	117. **A**	118. **D**	119. **B**	120. **B**
121. **D**	122. **D**	123. **B**	124. **C**	125. **A**
126. **C**	127. **C**	128. **C**	129. **A**	130. **D**
131. **D**	132. **A**	133. **B**	134. **D**	135. **D**
136. **C**	137. **A**	138. **B**	139. **D**	140. **C**

Numerical

141. **75**	142. **190 or 210**	143. **240**	144. **2**	145. **3**

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146. **6**	147. **3**	148. **36**	149. **12**	150. **2**
151. **2**	152. **3**	153. **30**	154. **15**	155. **3**
156. **10**	157. **30**	158. **5**	159. **21**	160. **12**
161. **492**	162. **500**	163. **3.33**	164. **25**	165. **25**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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